

**FEDERAL HIGHWAY ADMINISTRATION
TURNER-FAIRBANK HIGHWAY RESEARCH CENTER REPORT**

I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation:

Adhesive Anchor Sustained Load Testing Results

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CHAPTER 1. INTRODUCTION

On July 10, 2006 the ceiling system of the eastbound I-90 Seaport tunnel in Boston, Massachusetts partially collapsed. In support of the National Transportation Safety Board's (NTSB) investigation of this failure, the Federal Highway Administration (FHWA) dispatched a team of engineers and technicians from the Turner-Fairbank Highway Research Center (TFHRC) to conduct static tension load testing and load-displacement testing, and sustained load testing of the adhesive anchor rods used to support the ceiling system. The sustained load testing that was conducted on 11 anchor rods between July 26, 2006 and November 29, 2006, and both the load and load-displacement testing conducted on 15 anchor rods December 18, 2006 are the subjects of this report.

Observations made during a visual inspection of the adhesive anchors supporting the ceiling system shortly after the July 10, 2006 collapse revealed many anchors that had significantly displaced downward from their original embedment location. In response to these observations, TFHRC engineers attached dead-weights to two anchor rods in the westbound tunnel on July 26, 2006 and monitored their displacement response for nearly two days. These initial sustained load tests, referred to herein as the screen tests, were intended to indicate whether a more sophisticated investigation of the response of the anchor system to sustained load was needed. In fact, displacement of the screen test anchors never stabilized during the monitoring period and prompted the more rigorous investigation that is the primary focus of this report.

After the conclusion of the sustained load testing, four of the subject adhesive anchors that remained in the tunnel ceiling were removed with load-displacement testing conducted in full or substantial compliance with the static tension testing provisions of ASTM E 488 *Standard Test Method for Strength of Anchors in Concrete and Masonry Elements* (hereafter referred as the ASTM Specification). In addition to these sustained load test anchors, 11 anchors from elsewhere in the westbound tunnel were also load or load-displacement tested using the ASTM Specifications. Unless stated otherwise herein, the equipment and procedures used to conduct these tests are described in a previous report.¹

This report summarizes the equipment and procedures used, and the data produced during the screen tests and the more rigorous sustained load testing performed by the FHWA team on the aforementioned dates. Also included in this report are the data produced during the load and load-displacement testing conducted December 18, 2006. The following chronology briefly describes the dates, principles and events of this testing.

- July 26-28, 2006. Sustained load screen tests initiated: Hartmann, Ocel, Wright, Fuchs, Adams

¹ Hartmann, J., Ocel, J., Shipp, J., Wright, W. and Fuchs, P., "*I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation: Adhesive Anchor Load and Load-Displacement Testing Results*," Turner-Fairbank Highway Research Center Report, Federal Highway Administration, February 2007.

- September 26-28, 2006. Sustained load Tests 1 through 6 initiated: Hartmann, Ocel, Fuchs, Adams
- October 11-12, 2006. Sustained load Tests 3B and 5B initiated: Hartmann, Fuchs
- October 25-26, 2006. Sustained load Test 3C initiated: Adams, Fuchs
- November 6-7, 2006. Unload/reload cycle conducted on sustained load tests 1, 2 and 6: Wright, Fuchs
- November 28-29, 2006. All sustained load tests concluded: Hartmann, Fuchs
- December 18, 2006. Load and load-displacement testing for selected anchors: Hartmann, Ocel

ADHESIVE ANCHOR IDENTIFYING NOMENCLATURE

An identifying nomenclature for the adhesive anchors was established by NTSB personnel. This nomenclature has been consistently used throughout the field investigation and testing, and is illustrated using the following example adhesive anchor descriptor:

EB082-M1-3/4/5-NE

Tunnel:.....*EB* in this example indicates the East Bound tunnel (*EBD* also used).

WB indicates the West Bound tunnel (*WBD* also used).

Panel:.....*082* in this example indicates the panel module (module numbers were taken from the contract plans).

EB panel numbers include *080*, *081*, *082* and *083*.

WB panel numbers include *081*, *082*, *083* and *084*.

Line:.....*M1* in this example indicates hanger line within the tunnel.

EB hanger lines were labeled *N* for northern, *M1* for northern middle, *M2* for southern middle and *S* for southern.

WB hanger lines include *N* for northern, *M* for middle and *S* for southern.

Location:.....Location *3/4/5* in the example indicates the hanger to which clevis and turnbuckle suspension rods 3, 4 and 5 (*3,4,5* or *3-4-5* also used) were attached. Hanger locations were assigned a label that identified which suspension rods were attached to them. Suspension rods were numbered by panel starting with 1 on the eastern side and incrementally progressing west along a hanger line.

Anchor:.....*NE* in the example indicates the northeastern most adhesive anchor at this location. Locations indicated by a single number (representing a single suspension rod) had a pair of adhesive anchors securing the hanger to the ceiling. Adhesive anchors at these locations were labeled *N* for northern and *S* for southern.

Locations indicated by more than a single number (as in our example) had four adhesive anchors securing the hanger to the ceiling. These adhesive anchors were labeled *NE* for northeastern, *SE* for southeastern, *NW* for northwestern and *SW* for southwestern.

CHAPTER 2. SUSTAINED LOAD TESTING

The assessment of adhesive anchor sustained load behavior is vital to the understanding of service life performance. While this investigative technique captures all forms of inelastic response displayed by the anchor system, it is intended to focus on the susceptibility of the adhesive to move or deform permanently in an attempt to relieve stress over time. This phenomenon is generally referred to as creep and is affected most by the parameters of load level and duration, and temperature.

SCREEN TESTS

Since the initial observations from the scene suggested that the anchor system failure included a time element (the collapse occurred approximately 6 years after the completion of the ceiling installation) two sustained load screen tests were conducted on adhesive anchors in the westbound tunnel. Each of these anchors were loaded with approximately 1 ton lead weight and monitored for two days with an analog dial gage and a linear variable differential transducer (LVDT) to capture any displacement.

The 1 ton lead weights were suspended from the anchors using standard rigging equipment and steel wire rope. Figure 1 shows a drawing of the general test set-up. Briefly, a threaded coupler is turned halfway onto the adhesive anchor protruding from the ceiling. An extension of (~15") 5/8" diameter threaded rod is screwed into the remainder of the threaded coupler. A steel cantilever plate is double-nutted as shown in the figure onto the threaded rod extension. The cantilever plate had rough dimensions of 8"x 1.5" x 0.5" with an 11/16" hole centered on one end. The bottom of the threaded rod extension was fitted with a jamb nut and a 5/8" D-ring which was used to attached the steel wire rope. The wire rope extended to a D-ring attached to a 3-foot long, 3/4" diameter threaded rod which was inserted through the 1" diameter center-hole of the pyramid shaped lead weight. The weight was suspended from the rod with a thick plate washer and a heavy-hex nut.

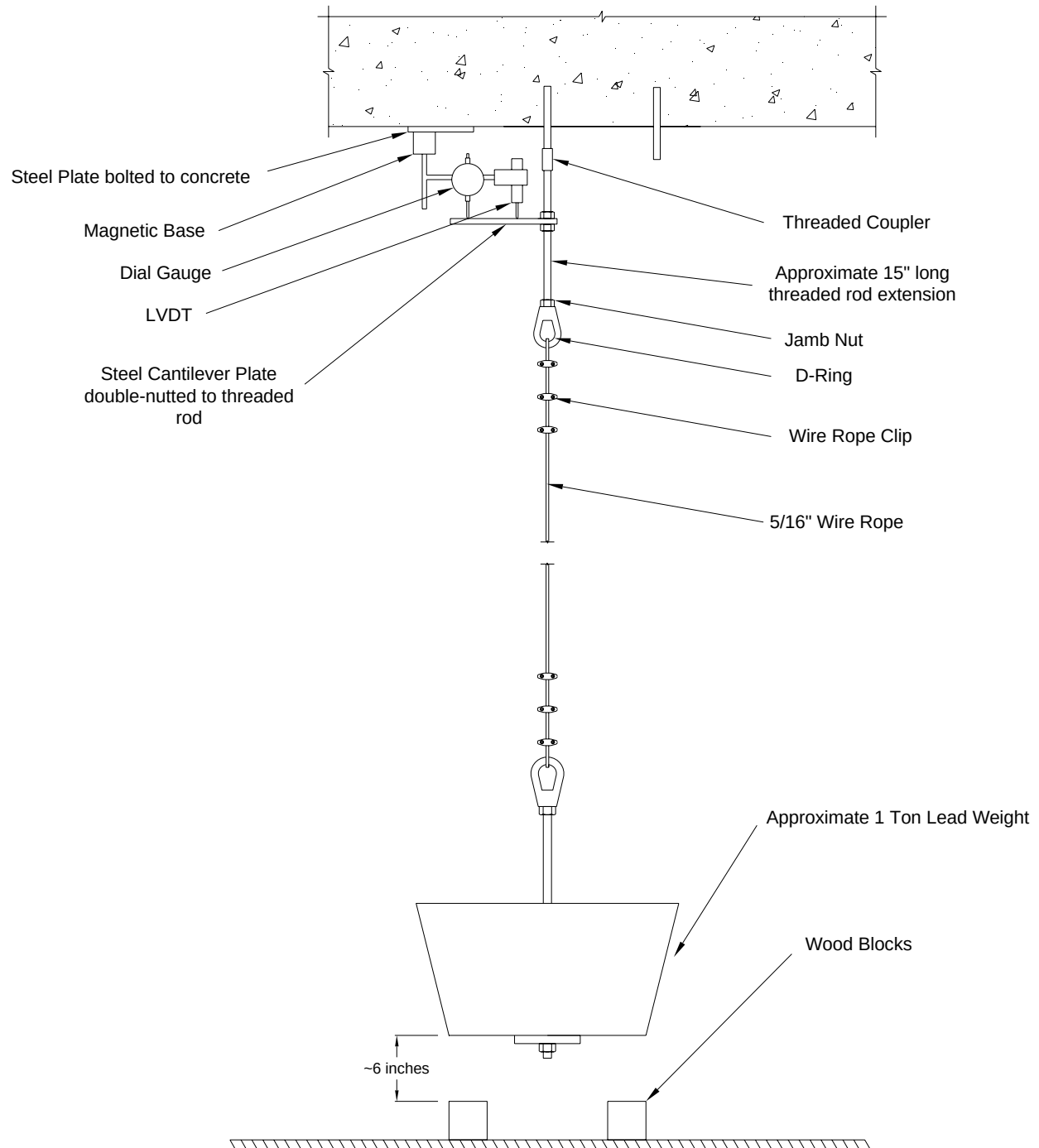


Figure 1 . Schematic of Sustained Load Test Setup.

The purpose of the cantilever plate was to create a flat, essentially horizontal surface that could be used by the instrumentation as a reference on which measurements could be based. A 6" x 6" x ½" steel plate was anchored to the concrete tunnel ceiling with a 3/8" concrete expansion anchor

to suspend the displacement monitoring instrumentation. Once the plate was in place, a magnetic base with articulating arms was used to position both the dial gage and LVDT. Two instruments are used to monitor displacement so that there is redundancy in the measurement. A picture of the instrumentation used can be seen in the right-side photo of Figure 2. The instrumentation setup shown in the figure which was used for the screen tests varied from that planned for these tests and shown on the drawing in Figure 1. Essentially the variation consists of the instrumentation monitoring displacement of the anchor from the underside of the cantilever plate instead of the top side of the cantilever plate. This change in setup was necessary to accommodate the clearance between the cantilever plate and the tunnel ceiling that resulted from an effort to minimize cyclic displacements in the cantilevered plate generated by any swinging of the lead weight caused by the tunnel ventilation system, wind or any other source.

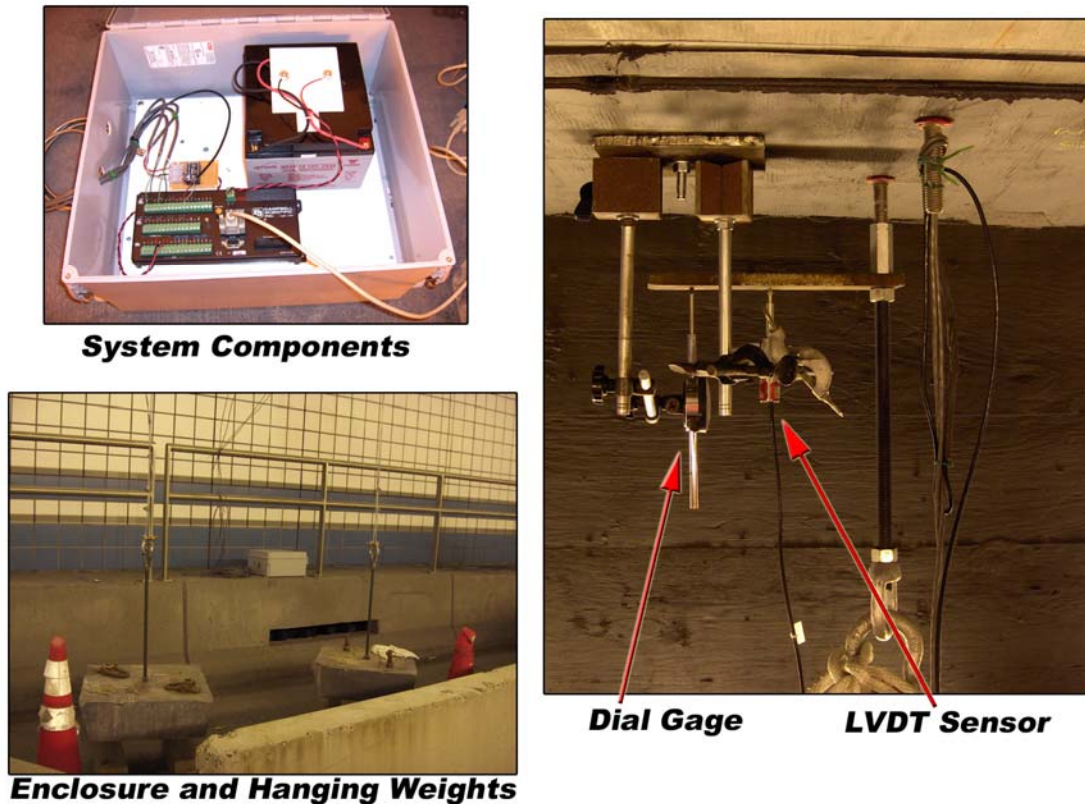


Figure 2 . Instrumentation Setup for Sustained Load Screen Tests.

While the analog dial gages were read visually, the LVDTs were interrogated at a rate of 0.2 Hz (1 measurement every 5 seconds) with a Campbell Scientific Model CR1000 datalogger (S/N: 2103) that was battery powered for the duration of the two-day screen tests. The data logger and battery were stored in a NEMA enclosure. From top to bottom, the photographs on the left side of Figure 2 show the datalogger and battery in the NEMA enclosure and the enclosure situated

on site behind the suspended lead weights. The LVDTs used were Model DCTH400AG manufactured by RDP Electrosense. Anchor WBD083-S-10-S was monitored with LVDT S/N: 65590 and anchor WBD082-1-S-1 was monitored with LVDT S/N: 67813.

A forklift was used to initially apply the load of the lead weights to each anchor. The forklift supported each of the lead weights from underneath approximately 12 inches above the roadway surface while the wire rope was attached to the D-ring described above. To transfer the load from the forklift to the wire rope and, ultimately to the adhesive anchor, the forks were slowly lowered from beneath the lead weight. Once the wire rope became taut, the weight was left suspended approximately 10 inches above the roadway. Displacement measurements began prior to the weight being suspended. Wooden cribbing was also placed below the lead weights so that if the anchor did displace completely from the ceiling the lead weight would not come to rest on the roadway. Figure 3 shows a picture of the screen tests in progress.



Figure 3. Sustained Load Screen Tests in Progress.

A summary of the screen test parameters are reported in Table 1. The two anchors that were the subject of these tests were subsequently load-displacement tested using the incremental method. The results of the load displacement tests were reported previously.¹

Table 1. Summary of Sustained Load Screen Test Parameters.

Anchor Label	Test Designation	Initial Protrusion	Approximate Sustained Load	Duration of Loading
		(inches)	(lbs)	(hours)
WBD083-S-10-S	Screen test	3.063	2000	47
WBD082-S-1-S	Screen test	2.969	2000	47

The displacement versus time history of anchors WBD083-S-10-S and WBD082-S-1-S are shown in Figures 4 and 5 respectively. The displacement shown is the total displacement which includes the initial elastic response in addition to the creep response. The figures show that the data from both the dial gage and LVDT are in agreement. This data has been adjusted to correct for the effects of bending in the plane of the instrumentation.

As shown on the figures, neither anchor's displacement stabilized during the approximately 47 hour duration of loading. Consideration of these results and that the anchors displayed no distress or displacement prior to this testing prompted the more rigorous sustained load investigation detailed in the next section.

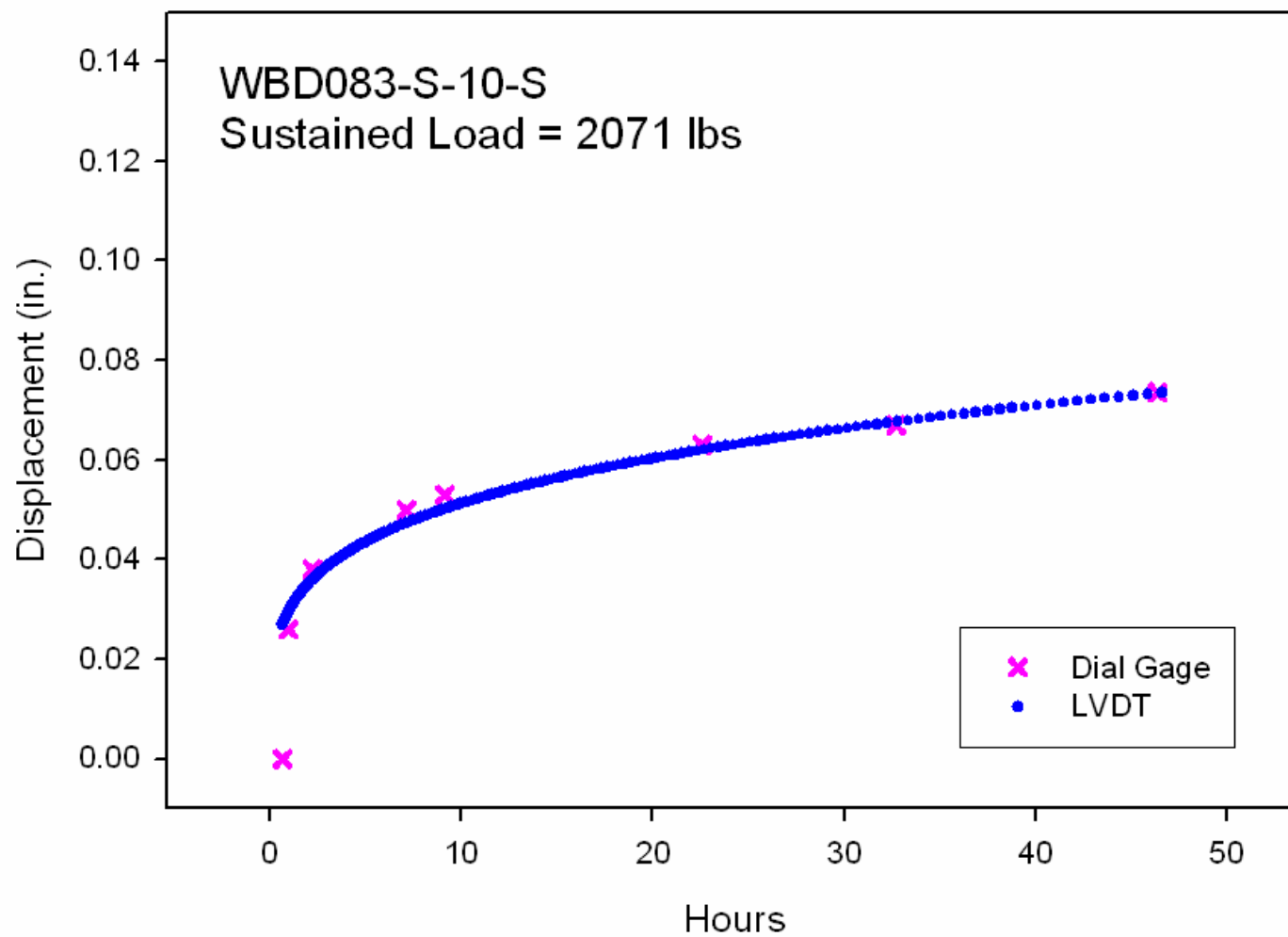


Figure 4. Sustained Load Behavior for Adhesive Anchor WBD083-S-10-S.

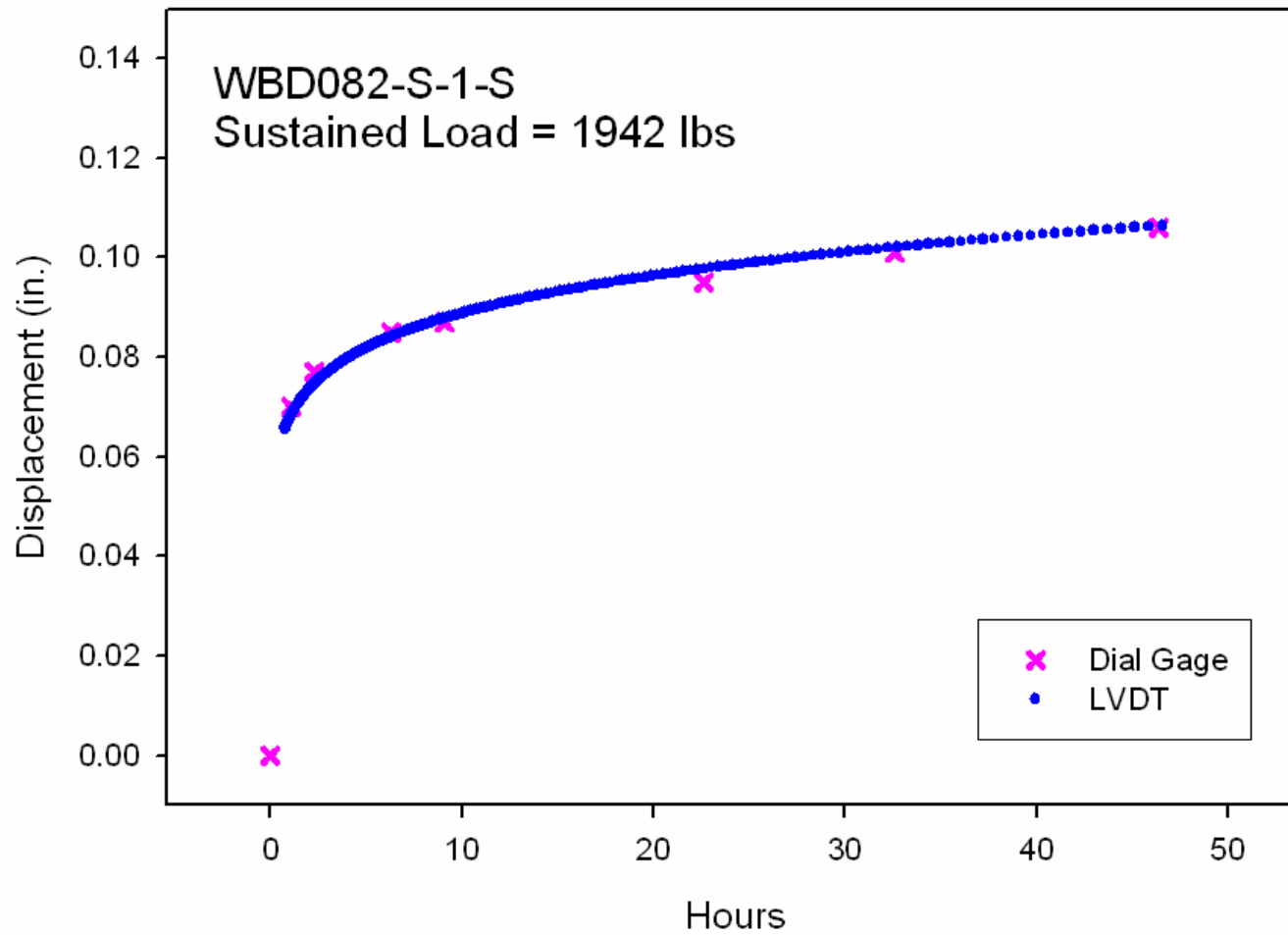


Figure 5. Sustained Load Behavior for Adhesive Anchor WBD082-S-1-S.

RIGOROUS SUSTAINED LOAD INVESTIGATION

In order to further characterize and quantify the time-dependent behavior observed during the sustained load screen tests, a matrix of 3 load levels that approximately represent the range of service loads experienced by the adhesive anchors was selected for more rigorous investigation. Concrete blocks were fabricated and sealed to impart loads of either 1000, 2000 or 3000 pounds to an individual anchor. Initially two replicates at each load level were planned for a total of six tests to be monitored for up to 3 months. However, a total of nine tests were conducted during this series of testing as a result of failure of anchors due to creep prior to the 3 month duration.

All sustained load tests were conducted on anchors that displayed neither signs of distress nor indication of pre-test displacement.

The instrumentation for the rigorous investigation was comprised of an analog dial gage, a digital dial gage and a LVDT. In contrast to the screen test setup, the instrumentation for the rigorous testing was attached directly to the adhesive anchor as shown in Figure 6. In the photograph on the right hand side of this figure labeled *Sensor Configuration*, the adhesive anchor (*Bolt*) is attached to an extension rod with a 3-inch threaded coupler. A cantilever plate double-nutted in place on the extension rod holds the instrumentation used to monitor movement of the anchor and air temperature. The instruments shown from left to right are the LVDT, the digital dial gage and the analog dial gage. In the photograph in the upper left the mounting location of the temperature sensor or thermocouple is shown.

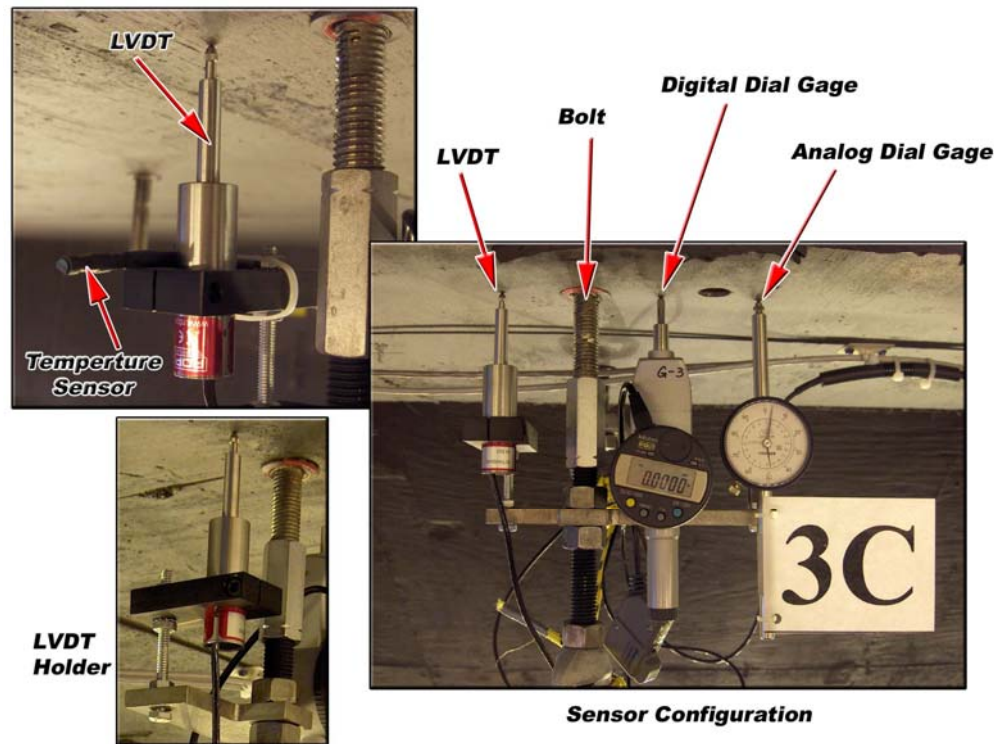


Figure 6. Instrumentation for Rigorous Sustained Load Testing.

The instrumentation setup for the rigorous sustained load testing was developed primarily to provide sufficient redundancy to insure consistency and correlation of the measurements or data produced. The LVDT was monitored at a rate of 2.5 Hz (5 measurements every 2 seconds) during events of high interest such as the initial loading and the unloading of anchors, and at a rate of 0.0017 Hz (1 measurement every 10 minutes) otherwise. Data was acquired using the same Campbell Scientific datalogger described for the screen tests. This measurement system was powered by battery which was recharged via a hardwired electricity source.

Two models of digital dial gage were used; Mitutoyo Digital Indicator Model No. 543-466B and Model No. 2424S-10. These gages were interrogated via a wireless system comprised of a L.S.Starrett Co. DataSure Gateway receiver (Model No. 1500-1-UN) and End Node transmitters (Model No. 1500-3-7N). The Gateway receiver utilized the wireless modem of a laptop computer that was secured onsite. Each of the digital dial gages had a separate transmitter. The digital dial gage measurements were collected at a rate of 0.1 Hz (1 measurement every 10 seconds) during events of high interest and at a rate of 0.0028 Hz (1 measurement every 6 minutes) otherwise. The dial gages and transmitters were powered by batteries with a life expectancy of several months. The computer was powered by internal battery which was recharged via a hardwire electricity source.

The analog dial gages were monitored by law enforcement personnel. In the absence of TFHRC staff, either an agent of the FBI or a Trooper of the MA State Police would visit the testing site

daily and record the measurements displayed on both the analog dial gage and digital dial gage face into a log.

This instrumentation setup proved to be very robust as high quality data was captured for each anchor throughout the duration of the 9 tests conducted. Also, the thermocouples and the orientation of the instruments permitted corrections to be made to the data for temperature change and bending about one-axis.

A summary of the sustained load test data is provided in Table 2. A photograph of the pre-test condition and any relevant additional data about the testing is included in the Test Specimen Register (Appendix A). Of the 9 anchors that were subjects of the rigorous sustained load investigation, 4 were subsequently strength tested with the incremental load-displacement method. Those results are reported in Chapter 3.

Table 2. Summary of Sustained Load Test Data

Anchor Label	Test Designation	Initial Protrusion	Approximate Sustained Load	Duration of Loading
		(inches)	(lbs)	(hours)
WBD084-S-4/5/6-NW	1	2.906	1000	1528
WBD084-S-8-N	2	3.000	2000	1528
WBD084-S-9-S	3	3.125	3000	377
WBD083-S-1-N	4	2.938	1000	1528
WBD083-S-3-S	5	3.063	2000	84
WBD083-S-4/5/6-NE	6	2.938	3000	1528
WBD083-S-2-N	3B	2.938	3000	7
WBD083-S-3-N	5B	3.094	2000	1149
WBD084-S-7-N	3C	3.250	3000	814

The concrete blocks fabricated to function as the loads for the sustained load tests were weighed both prior to testing in the TFHRC Structures Laboratory and upon test completion on-site in the tunnel. The results are shown in Table 3. The differences in these weight measurements are slight and attributed primarily to differences in the rigging used to suspend the blocks during weighing at each location and to damage incurred to the concrete. Several blocks had corners and edges knocked off during shipping, handling and installation.

Table 3. Summary of Dead-Weight Measurements

Test Block Designation		Pre-Test Weight (lbs)	Post-Test Weight (Nov. 28, 2006) (lbs)	Difference (lbs)
1	BY2	1028	1000	-28
2	MY2	2075	2052	-23
3	PX1	3071	3050	-21
4	BX1	1040	1025	-15
5	MX1	2030	2010	-20
6	PY2	3053	3033	-20

The displacement-time history of the nine rigorous sustained load tests are plotted in the same space in Figures 7 through 15. Figures 16 through 21 plot the records of Tests 3, 5 and 3B in space that best display behavioral characteristics.

Due to the sampling rate of the data acquisition systems, resolution of the instruments, the apparatus configuration and the process for the application of load, it is doubtful that the entire elastic component of anchor displacement is captured in any of the records of the sustained load test displacement-time histories. However, any displacement not captured in the total displacement records would be extremely minute and insignificant to behavioral interpretations.

Labels have been added in these test records to indicate several common features. These features are detailed in Table 4.

Table 4. Test Record Common Features

Feature Label	Description	Definition	Test Record
1	Unload/Reload	On November 6, 2006 the load was detached from the anchor and the anchor was monitored overnight. The load was then reapplied to the anchor on November 7, 2006. In all cases, the anchor behavior returned to the previous behavior curved.	1, 2, & 6
2	Unload & Recovery	On November 28, 2006 the sustained load testing was completed. All anchors still holding load were unloaded and monitored overnight to quantify recovery, if any, of the epoxy. Unload and recovery was monitored for Test 3 over 3 hours on October 12, 2006.	1, 2, 3, 4, 6, 5B & 3C
3	Swinging Load	Despite the best efforts of all involved, occasionally the hanging loads were ‘bumped’ or otherwise excited during visits of the investigative team, wind or tunnel ventilation system. The most significant of these events are labeled on the figures and were documented by investigators. Each of these events appears as an oscillation in the displacement versus time history of the effected anchor until the swinging action damps out or is stopped.	2, 6 & 3C
4	Dial Gage Out of Range	The dial gages had an approximate 2-inch gage length. This label indicates a reading that is most likely the result of the gage being out of range and not an actual measurement.	5 & 3B
5	LVDT Out of Range	The LVDTs had an approximate 0.9-inch gage length. This label indicates a reading that is most likely the result of the gage being out of range and not an actual measurement.	5 & 3B

All test records show significant agreement between the LVDT and dial gage data.

Figure 7 shows the displacement-time history for Test 1. This test applied approximately 1000 pounds to anchor WBD084-S-4/5/6-NW and lasted the duration of the rigorous sustained load testing investigation.

Figure 8 shows the displacement-time history for Test 2. This test applied approximately 2000 pounds to anchor WBD084-S-8-N and lasted for the duration of the rigorous sustained load testing investigation.

Figure 9 shows the displacement-time history for Test 3. This test applied approximately 3000 pounds to anchor WBD084-S-9-S and lasted about 377 hours. The test was terminated prior to the anchor displacing completely from the tunnel ceiling. Figure 16 shows the same data with the time scale reduced by a factor of approximately 30, and Figure 17 shows the data in its own space.

Figure 10 shows the displacement-time history for Test 4. This test applied approximately 1000 pounds to anchor WBD083-S-1-N and lasted for the duration of the rigorous sustained load testing investigation.

Figure 11 shows the displacement-time history for Test 5. This test applied approximately 2000 pounds to anchor WBD083-S-3-S and lasted about 84 hours. The test was terminated when the anchor displaced completely from the ceiling due to the sustained load. Figure 18 shows the same data with the time scale reduced by a factor of approximately 50, and Figure 19 shows the data in its own space.

Figure 12 shows the displacement-time history for Test 6. This test applied approximately 3000 pounds to anchor WBD083-S-4/5/6-NE and lasted for the duration of the rigorous sustained load testing investigation.

Figure 13 shows the displacement-time history for Test 3B. This test applied approximately 3000 pounds to anchor WBD083-S-2-N and lasted about 7 hours. The test was terminated when the anchor displaced completely from the ceiling due to the sustained load. Figure 20 shows the same data with the time scale reduced by a factor of approximately 200, and Figure 21 shows the data in its own space.

Figure 14 shows the displacement-time history for Test 5B. This test applied approximately 2000 pounds to anchor WBD083-S-3-N and lasted for about 1149 hours.

Figure 15 shows the displacement-time history for Test 3C. This test applied approximately 3000 pounds to anchor WBD084-S-7-N and lasted for about 814 hours.

Figure 22 shows the variation in ambient tunnel temperature recorded with the temperature sensors installed at anchors WBD084-S-8-N (Sensor 1) and WBD083-S-1-N (Sensor 2) throughout the duration of the rigorous sustained load testing. Sensor 1 was attached to the instrumentation monitoring sustained load Test 1 and Sensor 2 was attached to the instrumentation monitoring sustained load Test 4. The individual temperature measurements are presented in the figure as well as a linear regression trend line of daily averages. The horizontal bars shown indicate the duration limits of the individual sustained load tests.

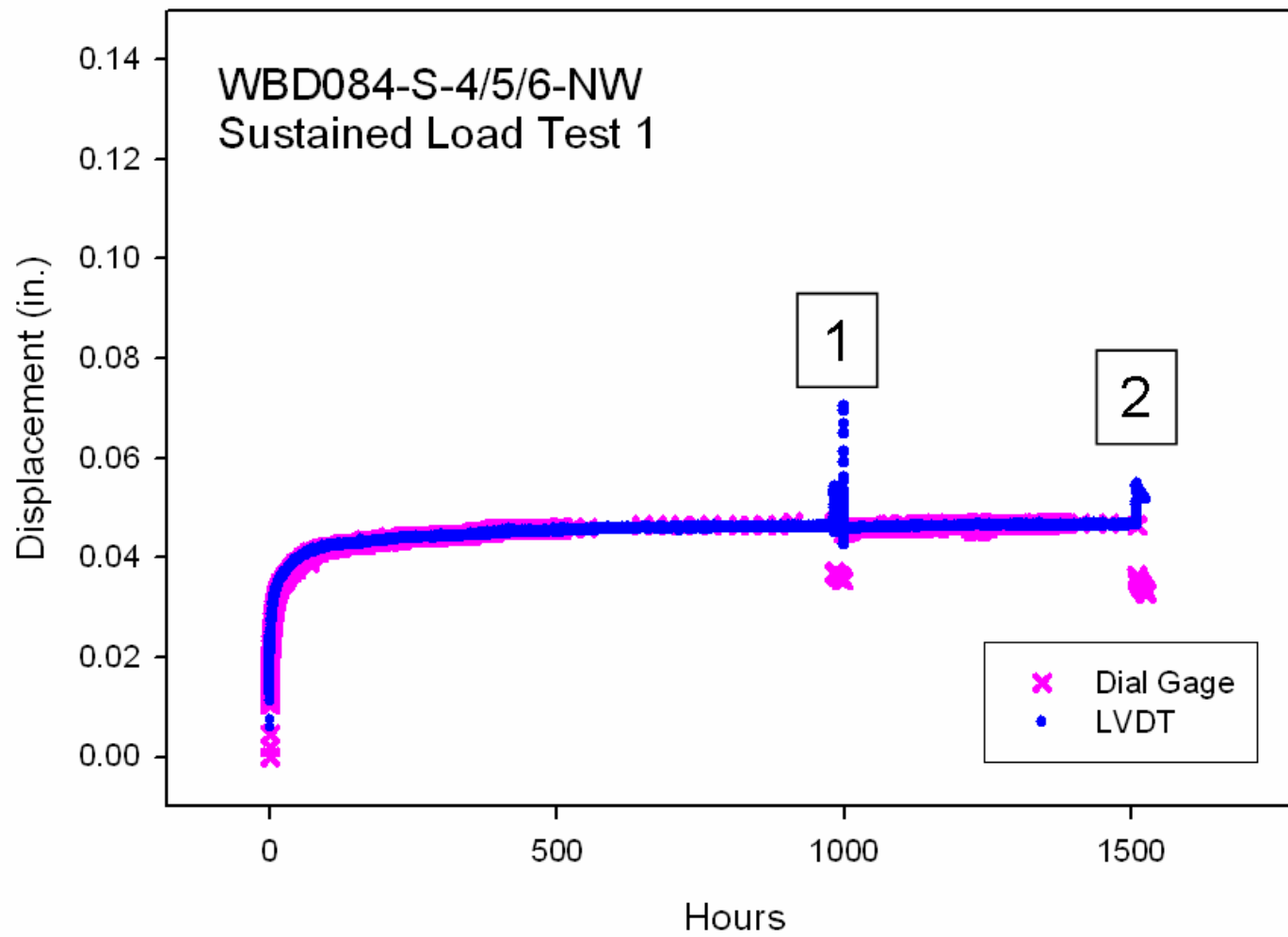


Figure 7. Sustained Load Behavior for Adhesive Anchor WBD084-S-4/5/6-NW.

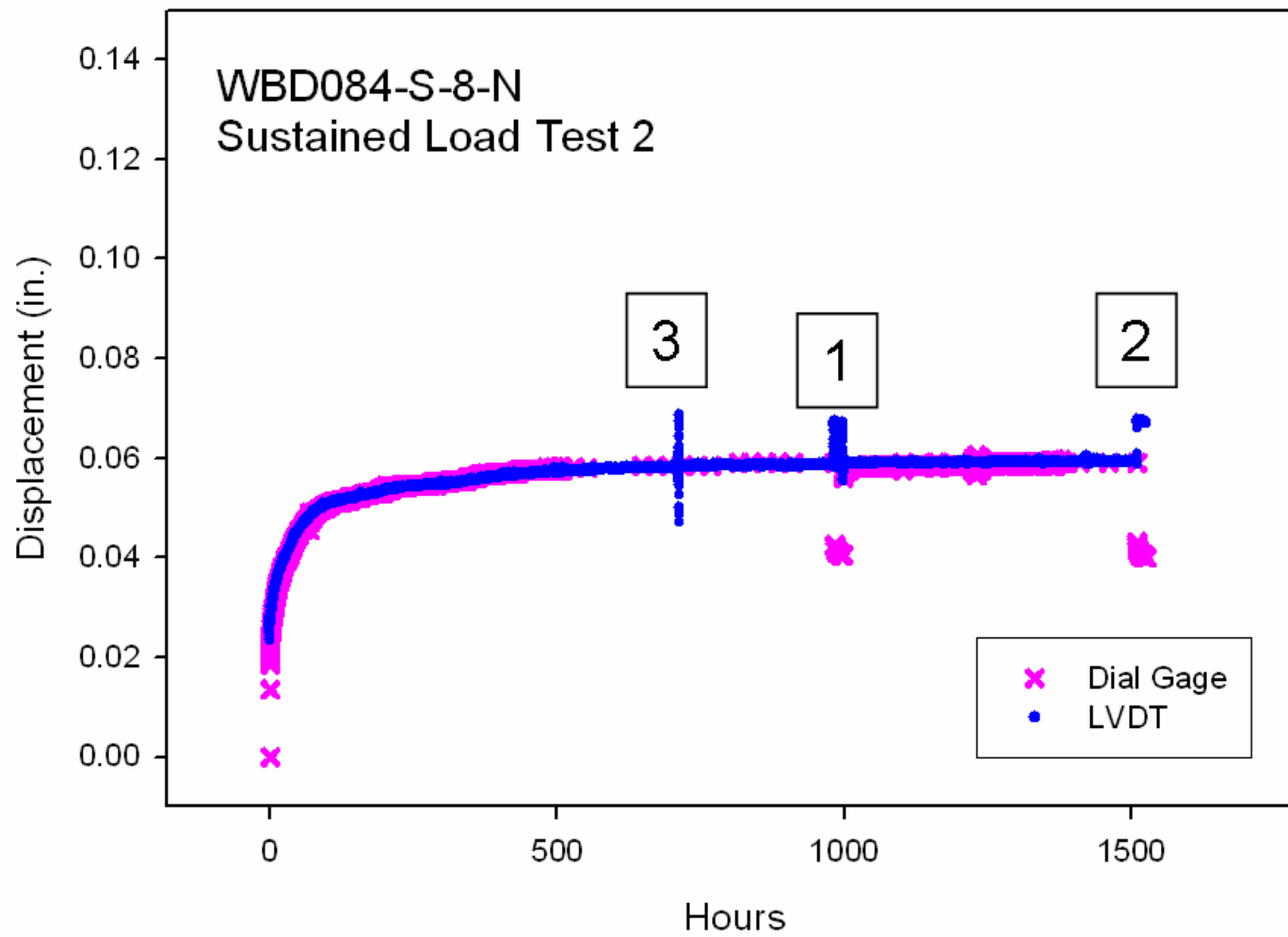


Figure 8. Sustained Load Behavior for Adhesive Anchor WBD084-S-8-N.

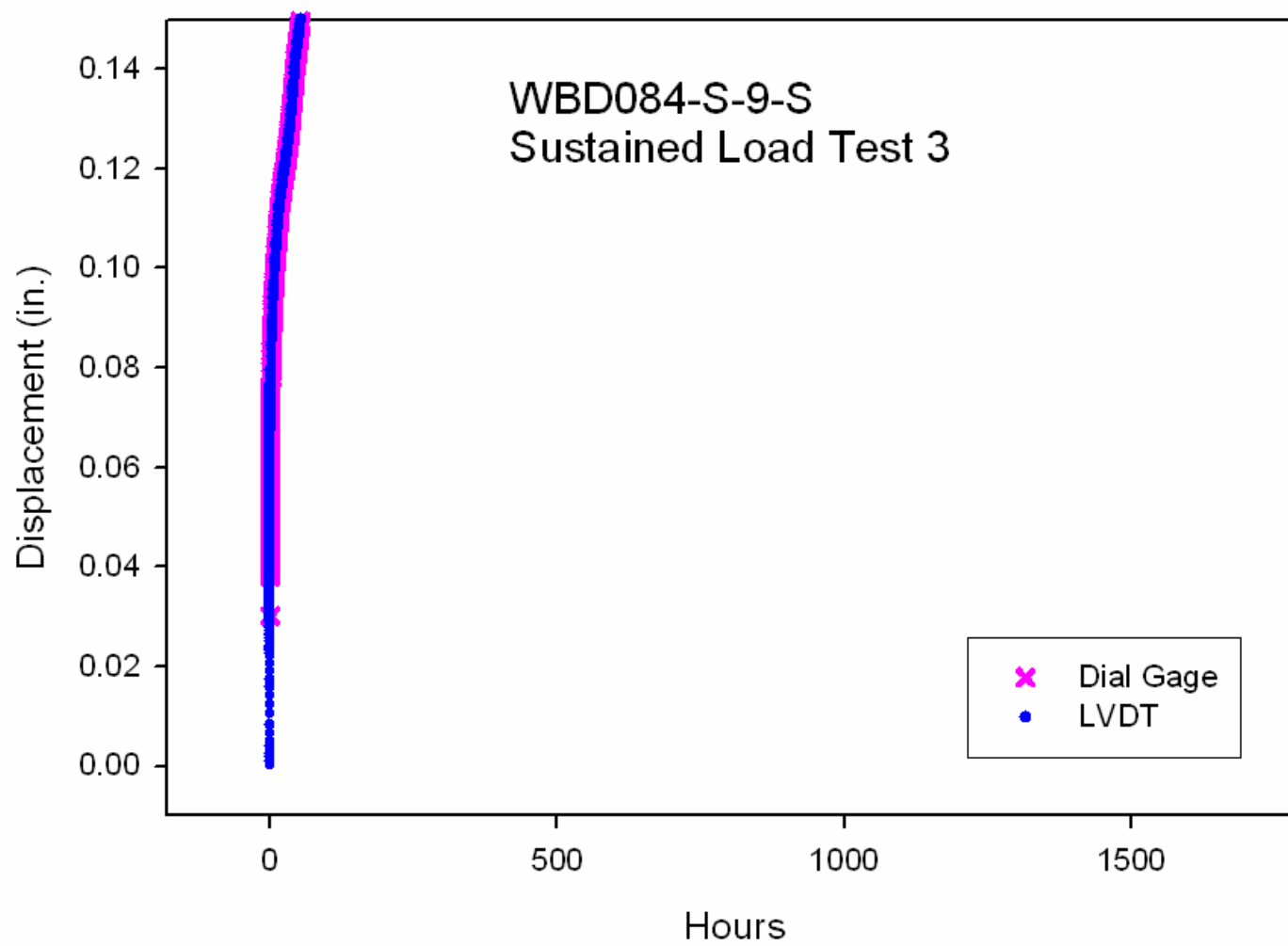


Figure 9. Sustained Load Behavior for Adhesive Anchor WBD084-S-9-S.

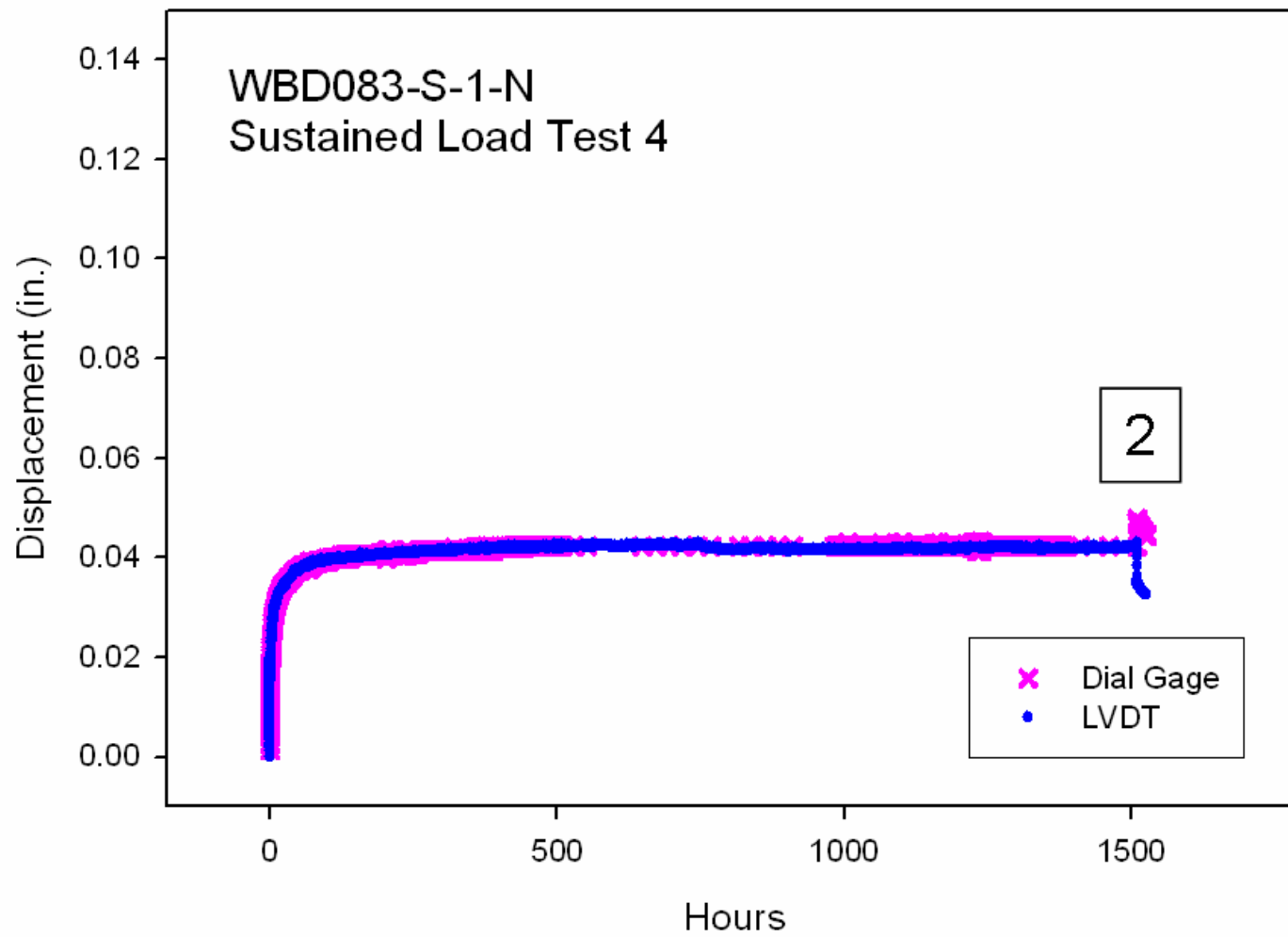


Figure 10. Sustained Load Behavior for Adhesive Anchor WBD083-S-1-N.

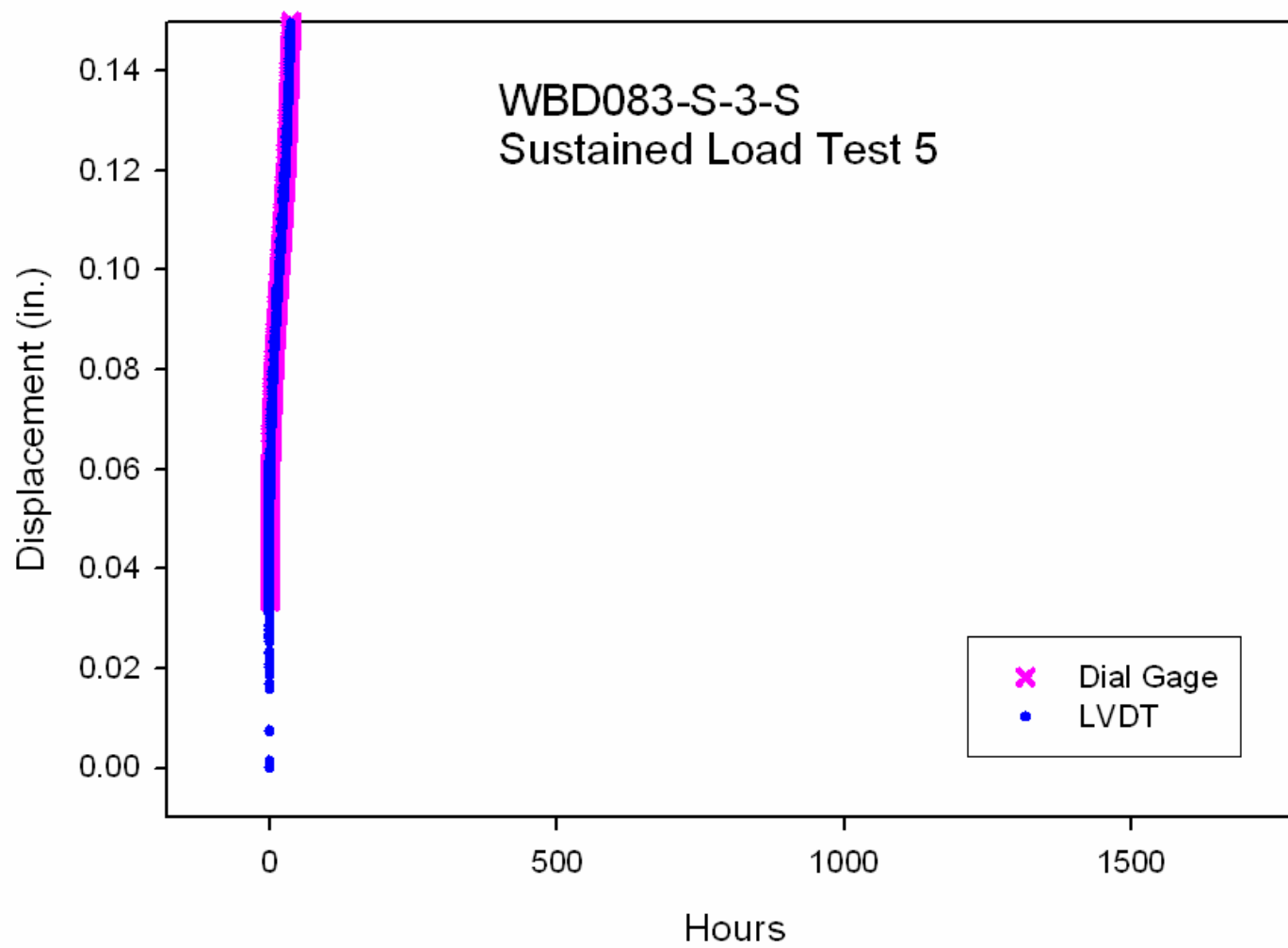


Figure 11. Sustained Load Behavior for Adhesive Anchor WBD083-S-3-S.

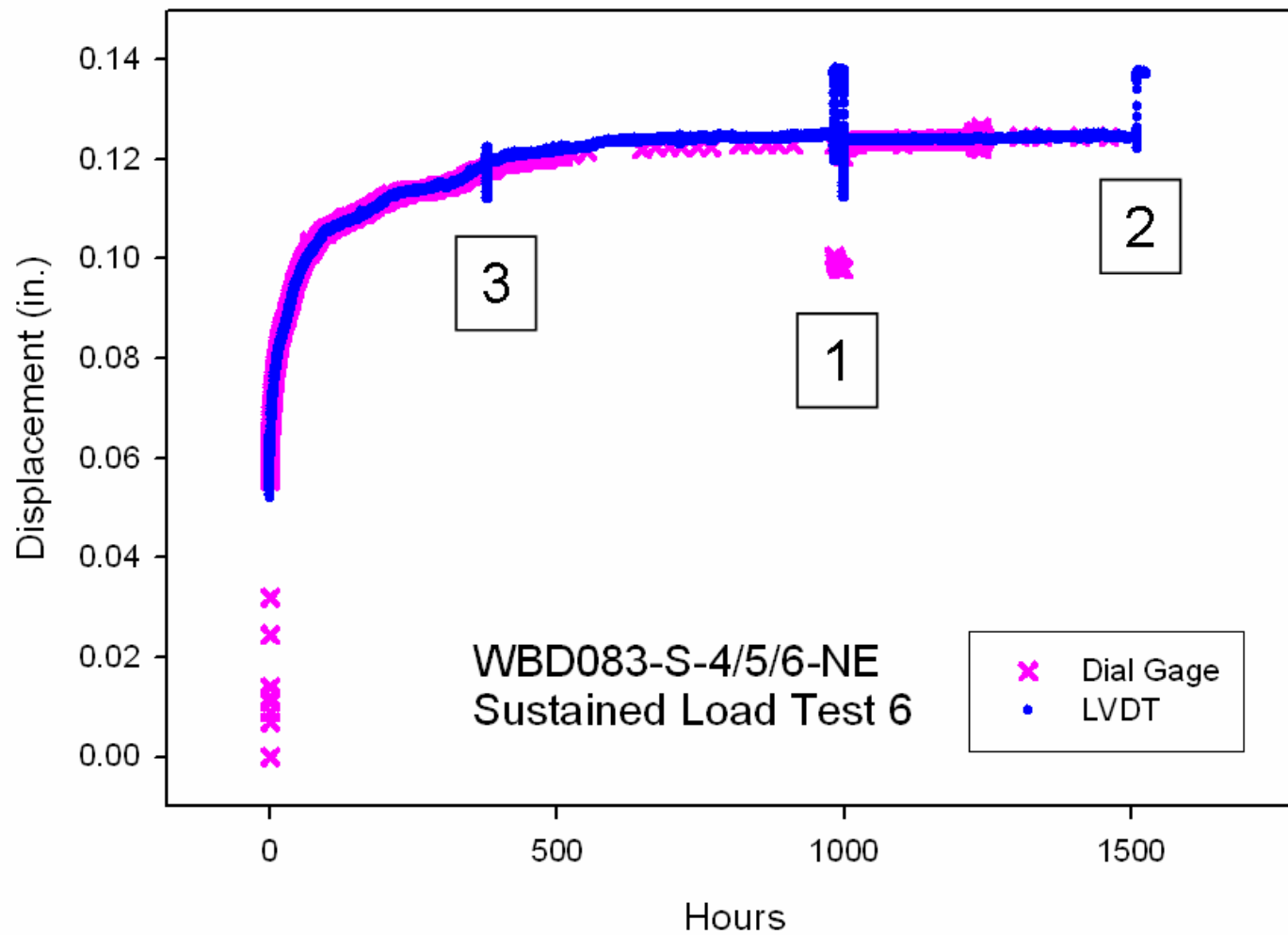


Figure 12. Sustained Load Behavior for Adhesive Anchor WBD083-S-4/5/6-NE.

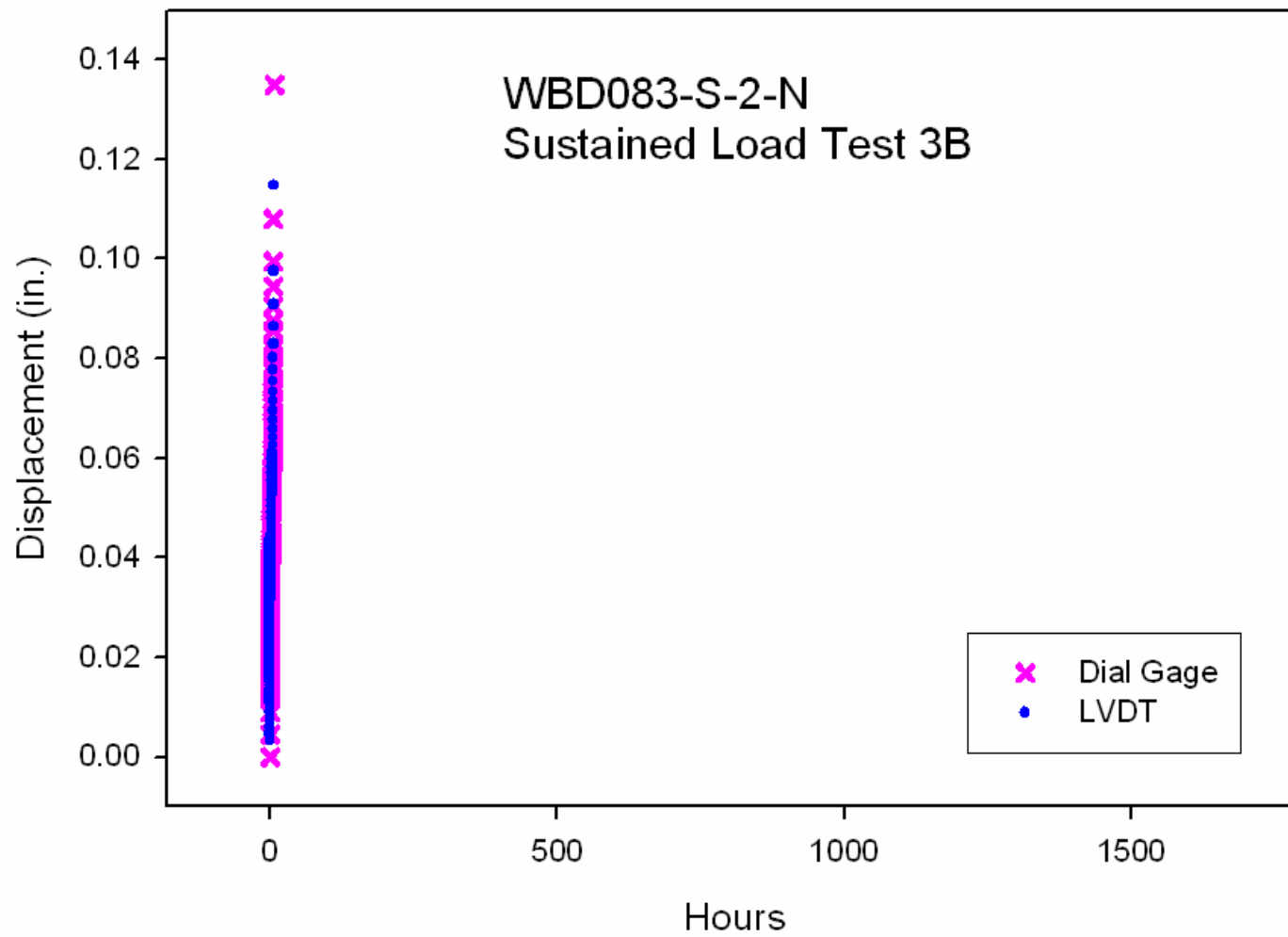


Figure 13. Sustained Load Behavior for Adhesive Anchor WBD083-S-2-N.

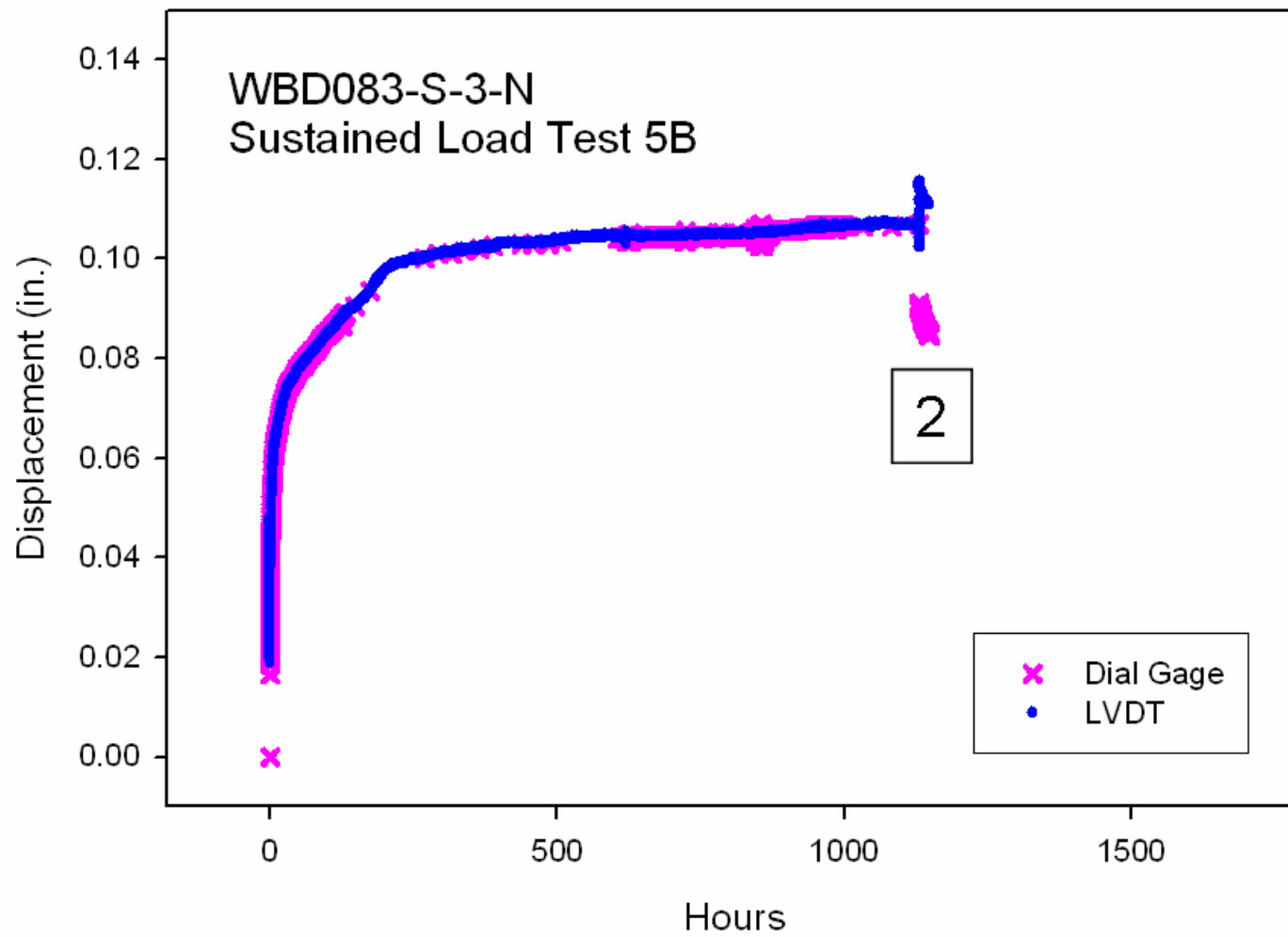


Figure 14. Sustained Load Behavior for Adhesive Anchor WBD083-S-3-N.

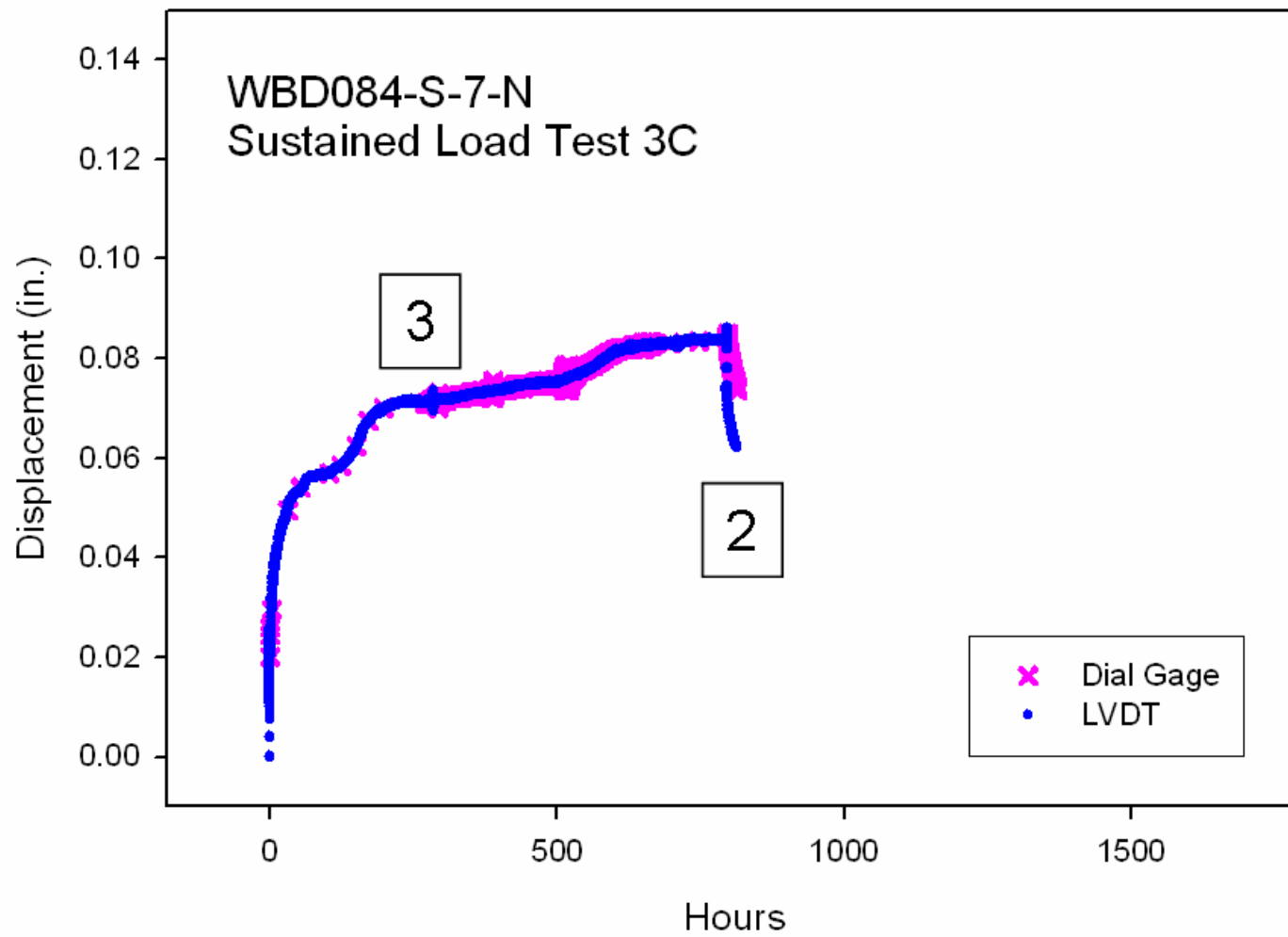


Figure 15. Sustained Load Behavior for Adhesive Anchor WBD084-S-7-N.

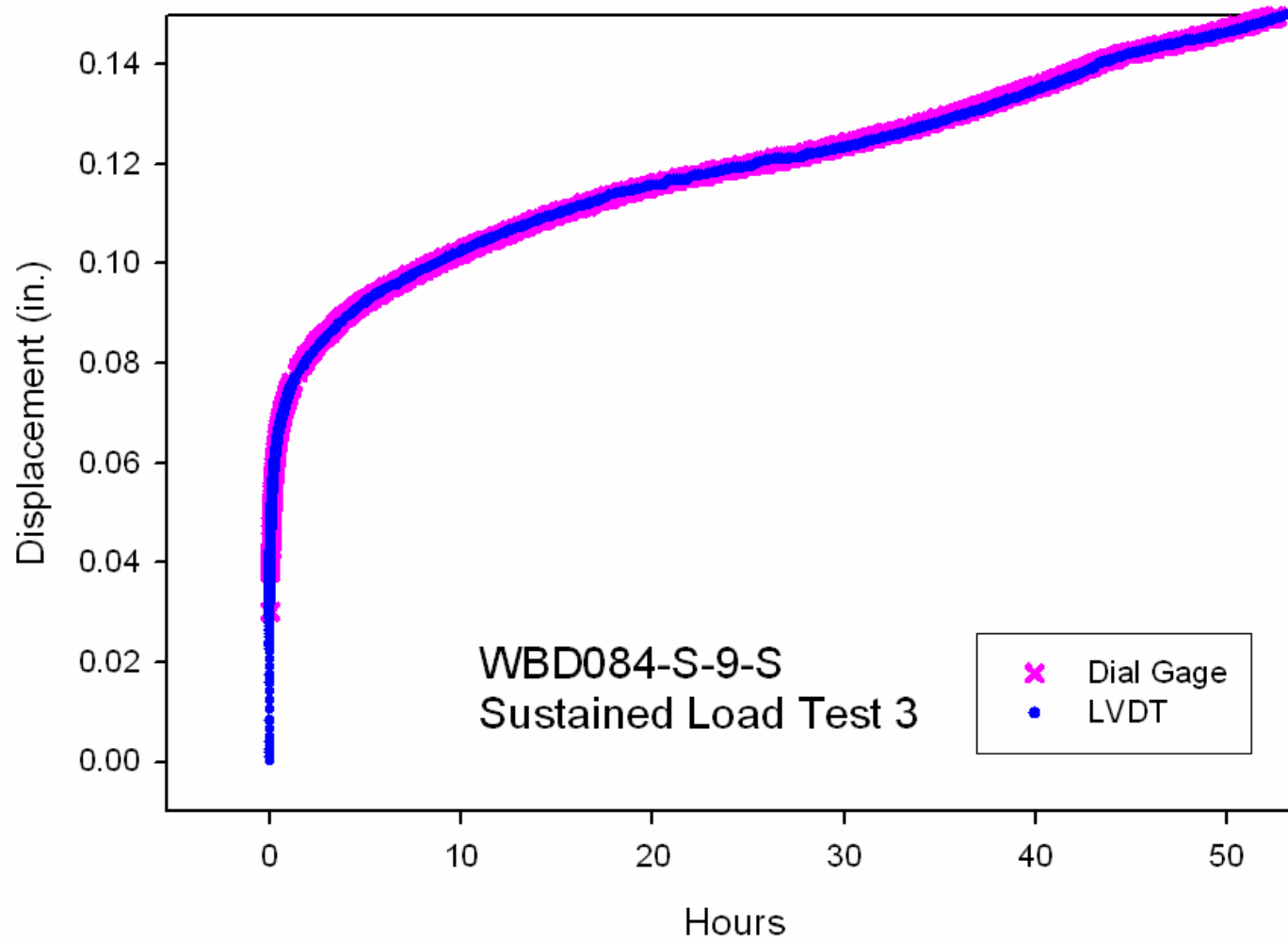


Figure 16. Sustained Load Behavior for Adhesive Anchor WBD084-S-9-S.

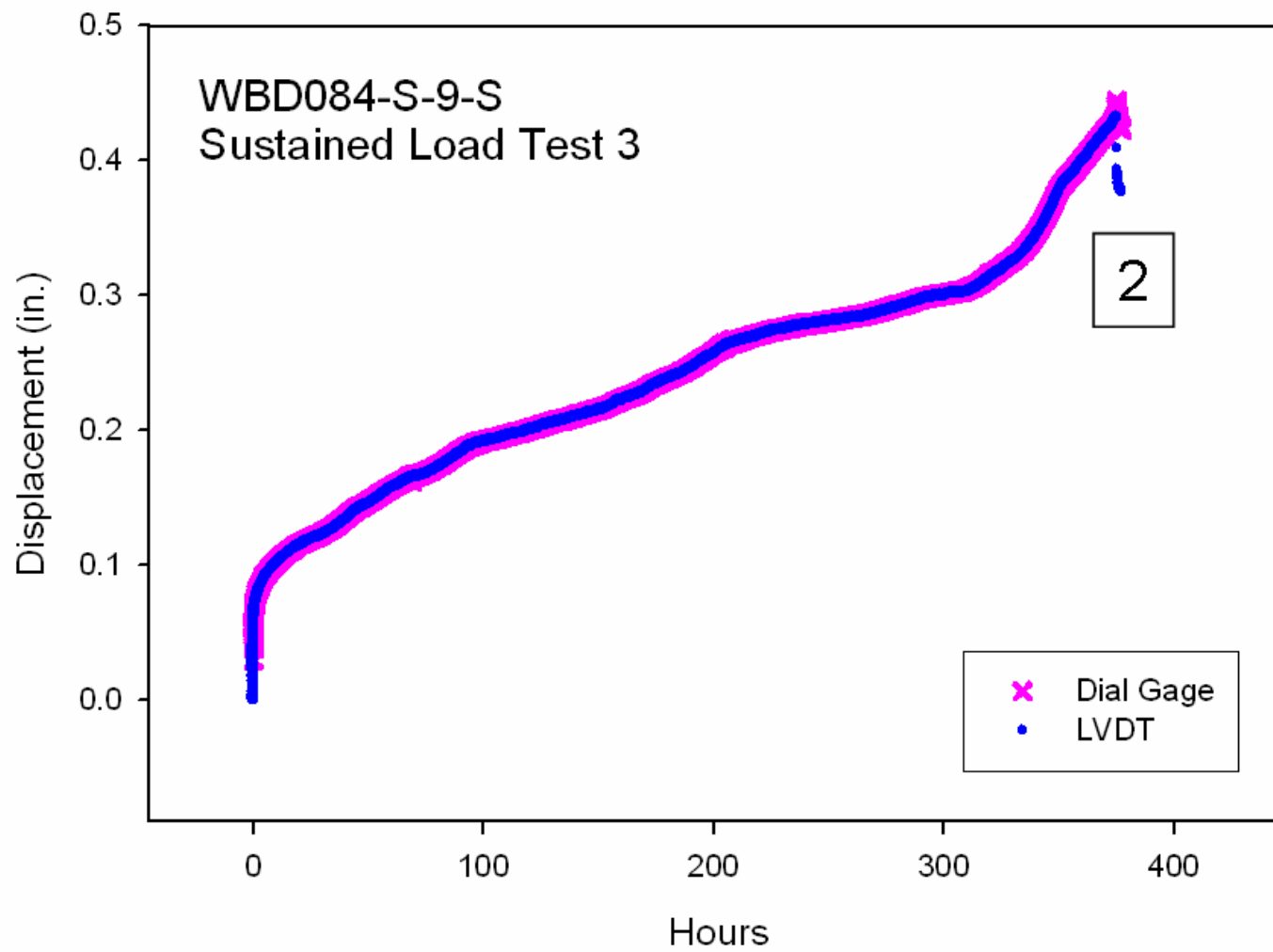


Figure 17. Sustained Load Behavior for Adhesive Anchor WBD084-S-9-S.

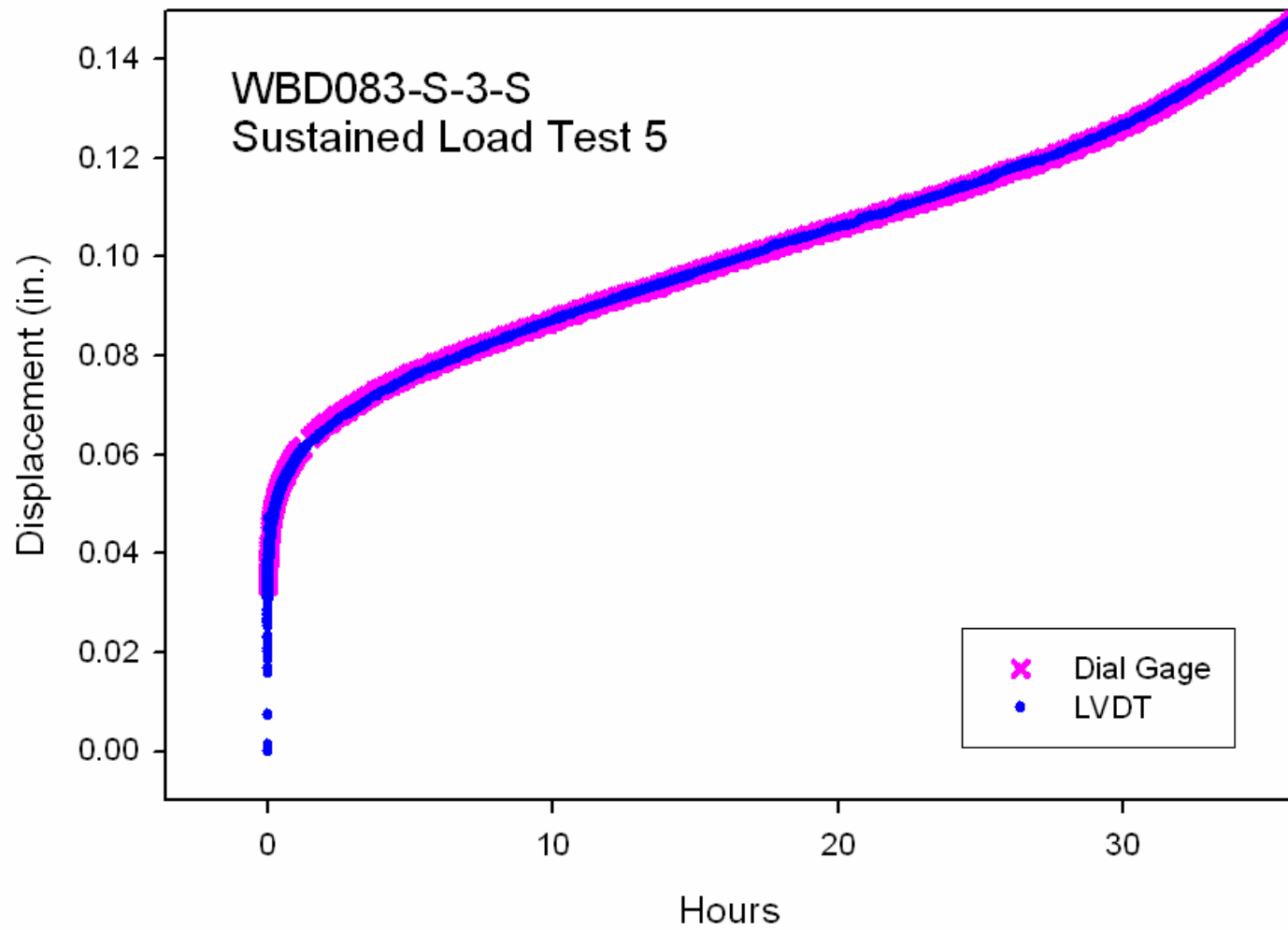


Figure 18. Sustained Load Behavior for Adhesive Anchor WBD083-S-3-S.

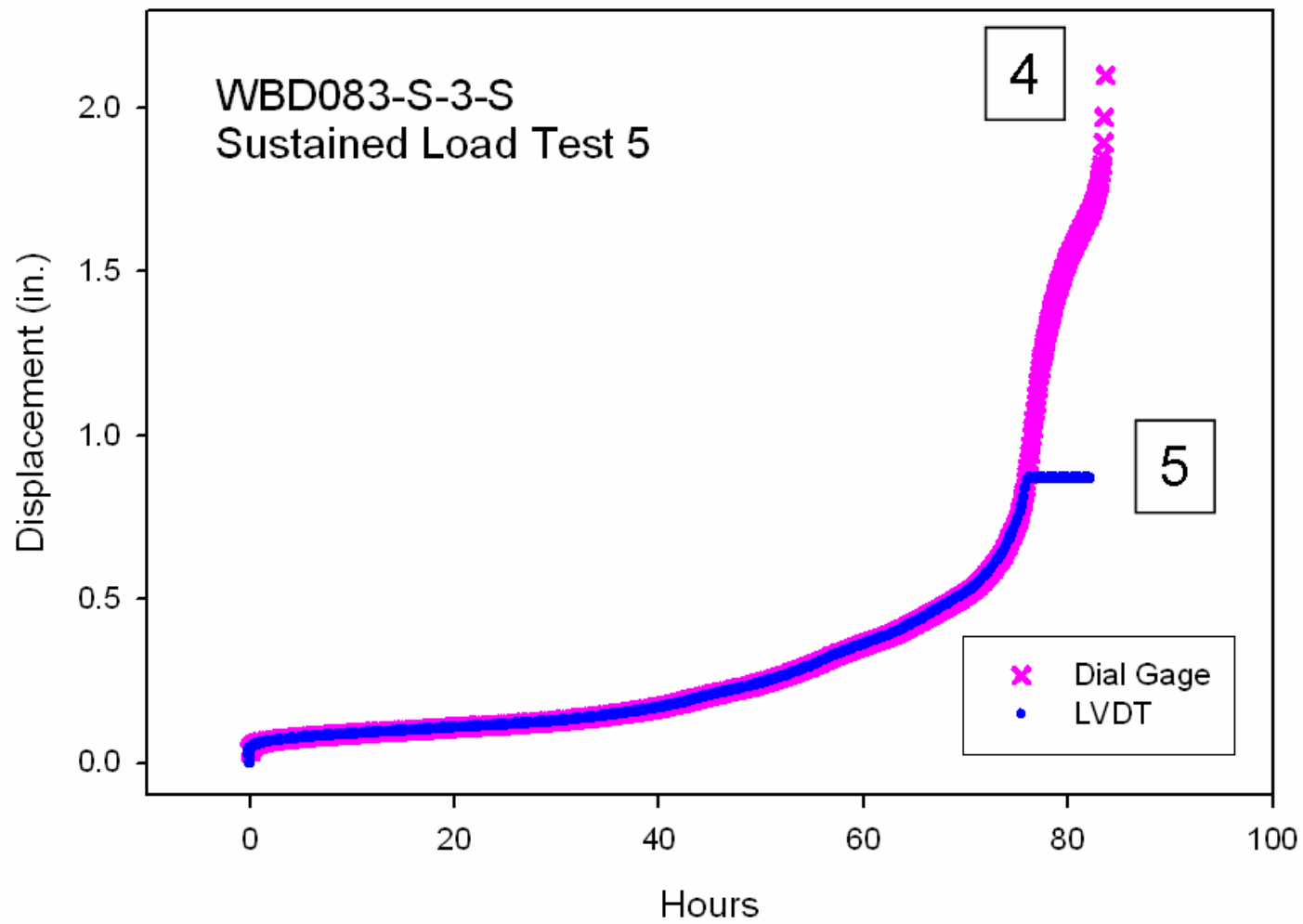


Figure 19. Sustained Load Behavior for Adhesive Anchor WBD083-S-3-S.

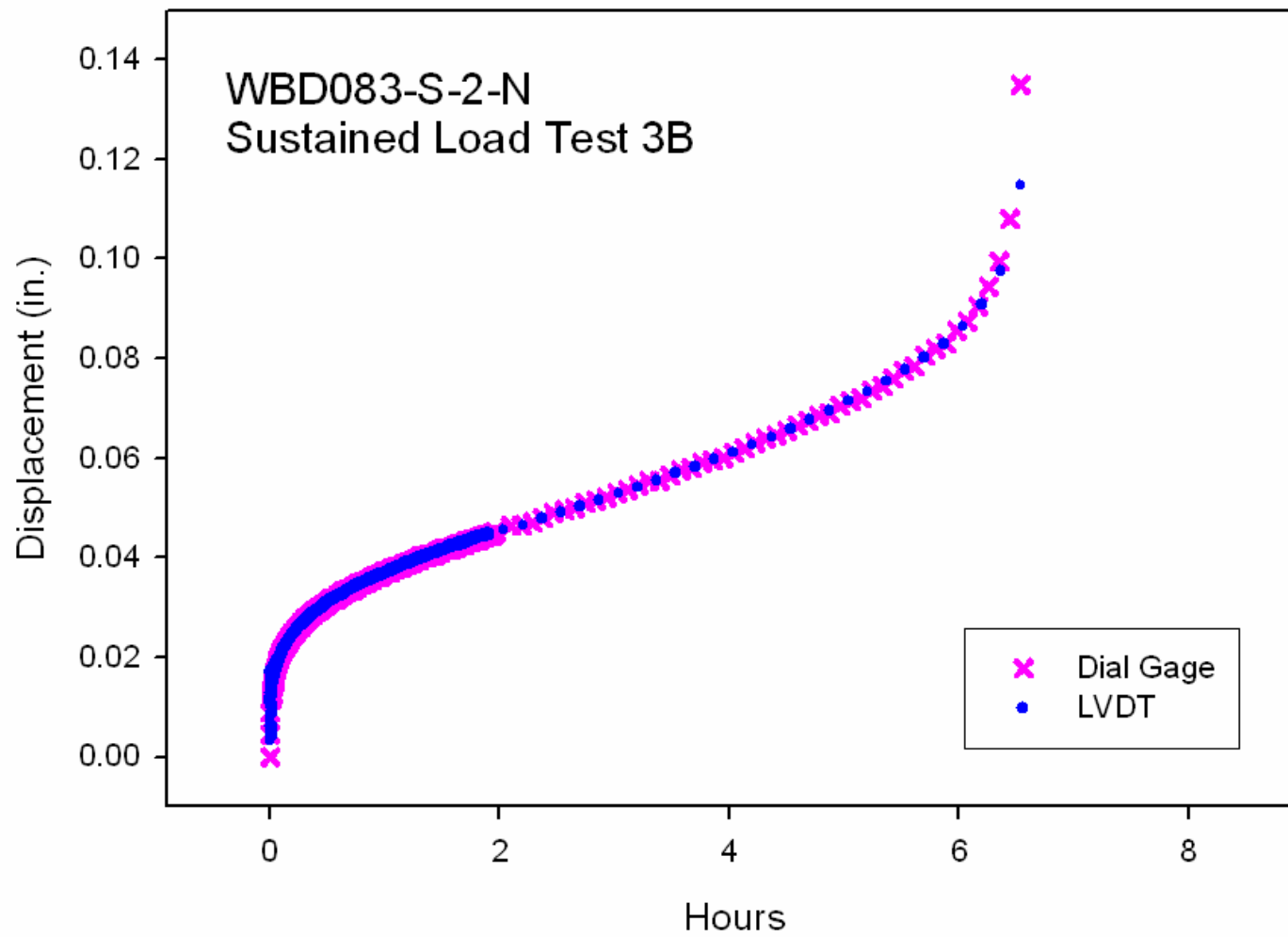


Figure 20. Sustained Load Behavior for Adhesive Anchor WBD083-S-2-N.

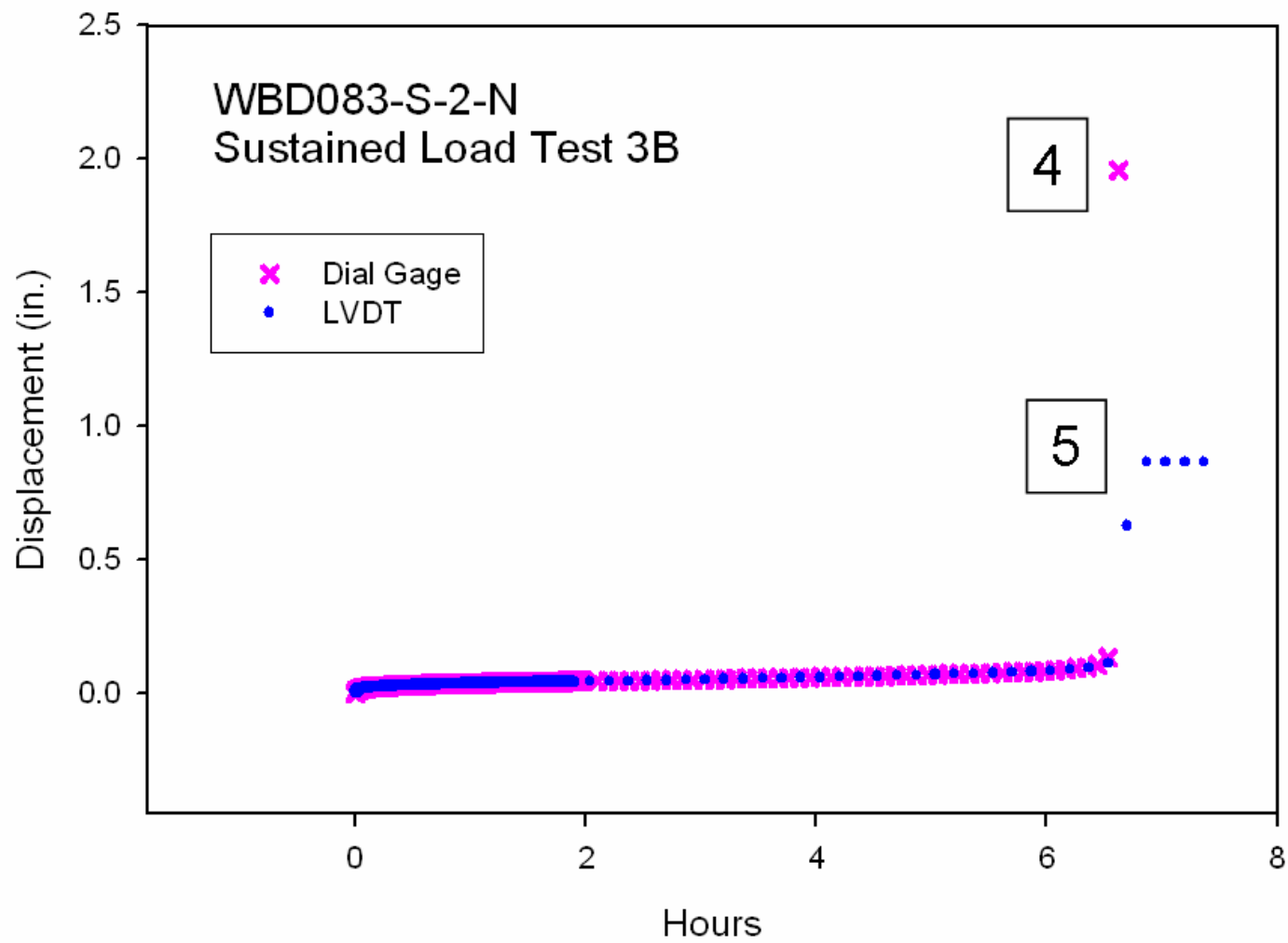


Figure 21. Sustained Load Behavior for Adhesive Anchor WBD083-S-2-N.

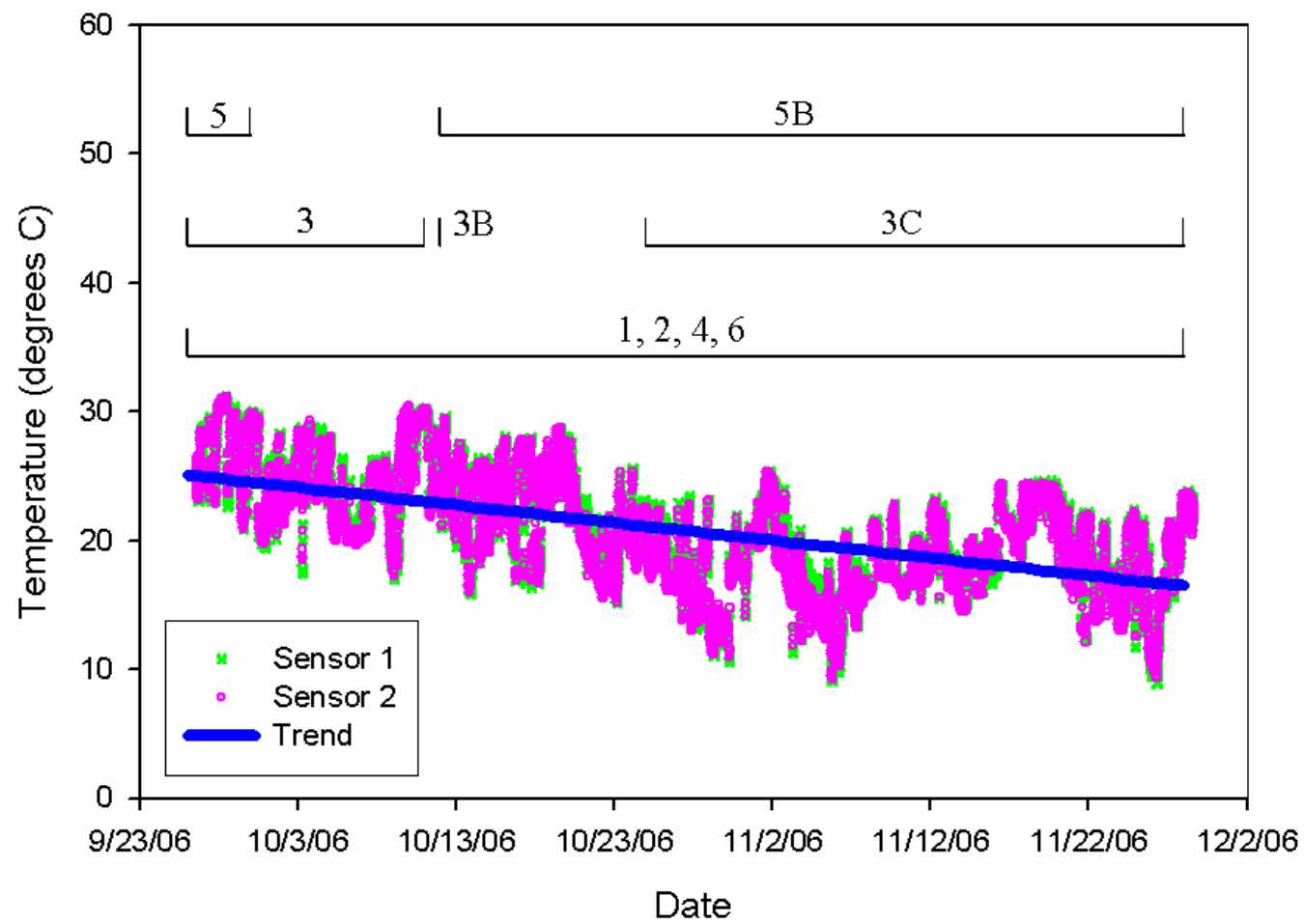


Figure 22. Tunnel Temperature Variation During Sustained Load Testing.

CHAPTER 3. LOAD TESTING

The load testing conducted on December 18, 2006 used the loading apparatus shown in Figure 23. This system employed two steel wide-flange sections and a “needle-beam” to react the jack off the ceiling and transfer load to the adhesive anchor. As the wide-flanges reacted off the ceiling at a distance from the anchor greater than the depth of anchor embedment, this system is used for unconfined testing. Unconfined testing is the preferred method as it allows failure to occur in the concrete and does not bias any failures occurring in the adhesive by imparting localized compressive stresses.

Referring to Figure 23, the W8 wide-flange sections were attached to the ceiling using 3/8” mechanical anchors and positioned such that there was a minimum clear distance of 11 inches between the side of the anchor rod and the flange tip of each wide-flange section. The needle-beam is comprised of two steel C-channel sections placed back-to-back but welded to end plates that space the webs approximately 2 inches apart. The needle-beam spanned between the two wide-flange sections and was held in place with C-clamps (not shown). Before it was clamped tight, the gap between the needle-beam webs was centered over an anchor rod. A coupler was used to attach an additional length of threaded rod passed through the web gap of the needle-beam to the in-situ adhesive anchor. Washers, the hollow core 12 ton Enerpac hydraulic jack and the Wagner Instruments 50-kip load cell were all centered on the threaded rod and held in place with a heavy hex nut. A hand pump was used to pressurize and extend the jack and a Data Translation Model 9804 USB Data Acquisition Module and laptop computer were used to monitor the load cell and capture the peak load resisted by the adhesive anchor prior to failure. The weight of the loading apparatus experienced by the adhesive anchor was approximately 20 lbs.

Two different needle-beams were used. In general, a needle-beam comprised of 4-inch deep C-channels was used to test anchor rods at locations where two anchors were present, and a needle-beam comprised of 6-inch deep C-channels was used to test anchor rods at locations where 4 anchors were present. The deeper channels were required to keep the system elastic while spanning the entire 4 anchor group and maintaining the necessary offsets for unconfined testing.

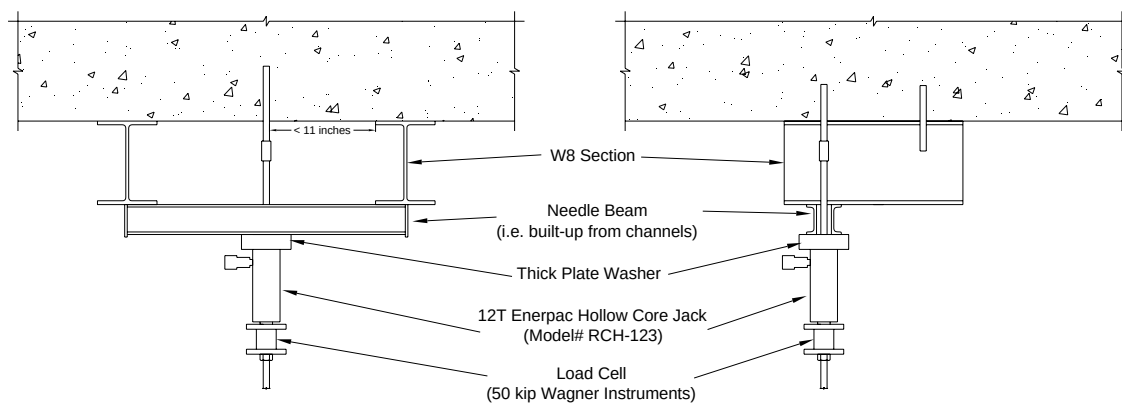


Figure 23. Loading Apparatus for Unconfined Testing.

LOAD TEST PROCEDURE

The procedures for used load testing on December 18, 2006 duplicated those used previously¹ with the exception that load and time were recorded electronically using a Data Translation model 9804 USB Data Acquisition Function Module and laptop computer. Briefly, the initial adhesive anchor in-situ protrusion was measured and photographed. The load apparatus was then connected to the adhesive anchor rod and the spacer plates, hydraulic jack, and load cell were all aligned. The load cell was balanced and zeroed, and then loading proceeded in accordance with the ASTM Specification.

Load was applied at an approximate rate of one stroke from the hand pump every 5 seconds. This typically caused the peak load to occur in about one minute. Once the peak load was attained, loading continued until a post failure peak load was attained, if possible. The post failure peak load was then recorded. Then the adhesive anchor was removed from the ceiling by continued jacking. Once out of the ceiling, custody of the adhesive anchor was taken by an accompanying crime scene investigator (FBI Special Agent or MA State Policeman). This process was consistently repeated for all load tests conducted December 18, 2006.

LOAD TEST RESULTS

Table 5 summarizes the initial protrusion, the time to peak load and the peak load measured during each load test reported. The data contained in this table summarizes that reported for each adhesive anchor individually in Appendix A. A pre-test photograph of the in-situ condition of each anchor is also included in Appendix A.

Initial protrusion measurements were made using an adjustable hand-square in fractions of an inch to the nearest 1/32 inches and reported in decimal form to 3 places. These measurements do not intend to report accuracy to 3 decimal places; rather the numbers are solely the result of a truncated conversion from fraction to decimal form. Initial protrusion measurements were made by placing the end of the hand-square rule against the tunnel ceiling and adjusting the headpiece so that the 90-degree leg rested on the end of the adhesive anchor as shown in the figures of Appendix A.

Table 5. Summary of Load Test Data

Anchor Label	Initial Protrusion (inches)	Peak Load (lbs)	Time to Peak Load (sec.)
WBD084-S-4/5/6-SE	2.813	24389	80
WBD084-S-8-S	3.531	15519	80
WBD083-S-4/5/6-NW	3.000	21688	53
WBD083-S-4/5/6-SE	2.844	21819	50
WBD083-S-9-N	2.875	17460	52
WBD083-S-9-S	2.938	23374	77

CHAPTER 4. LOAD-DISPLACEMENT TESTING

The load-displacement testing conducted on December 18, 2006 used the loading apparatus described in Chapter 3. Displacement measurements were made using three linear variable differential transformers (LVDT). For these measurements, the LVDTs were mounted to a circular aluminum plate which was double-nutted to the adhesive anchor as shown in Figure 24. The 7-inch diameter aluminum plate had a 0.75-inch hole in the middle. The three LVDTs were spaced evenly on a 6-inch diameter circle and were held into the aluminum plate with nylon thumb screws.

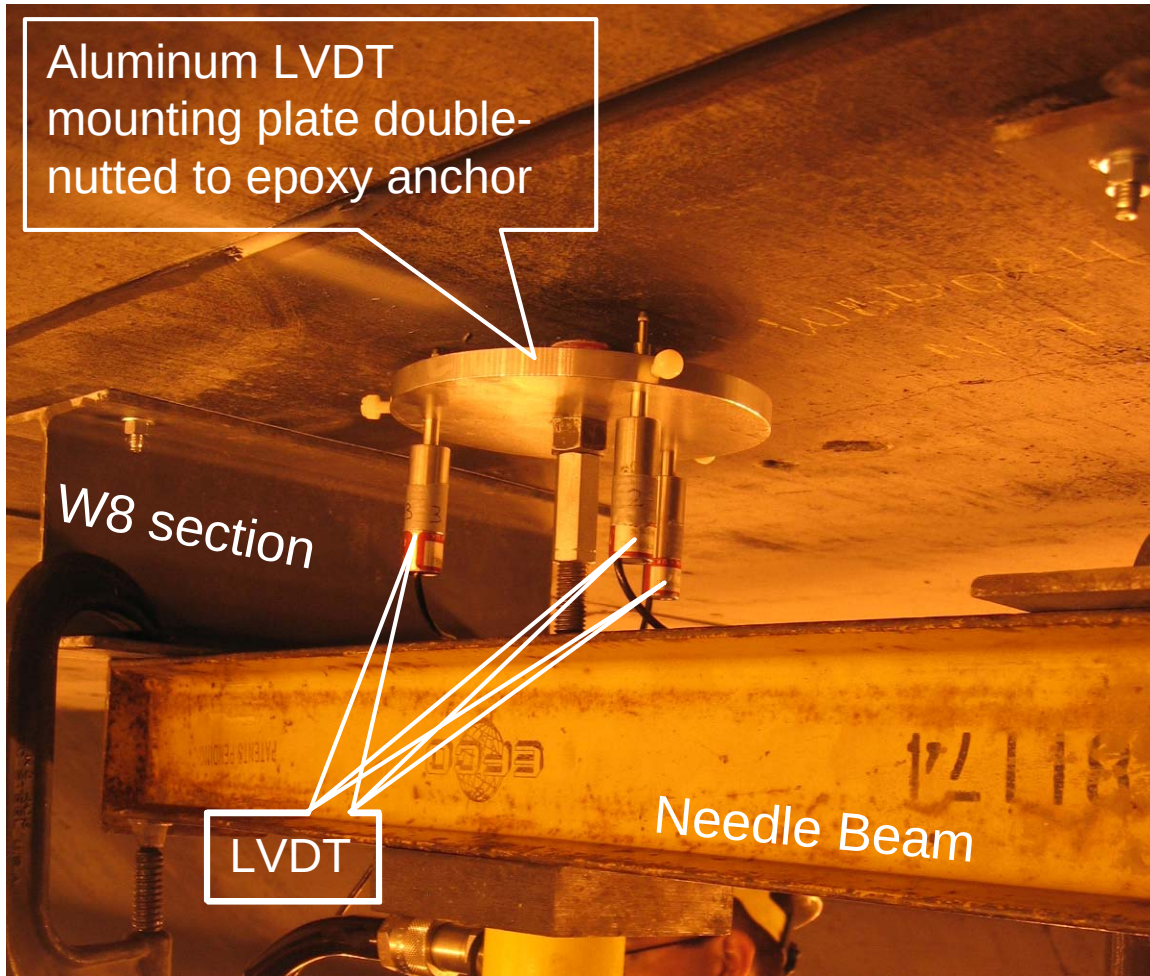


Figure 24. LVDT Mounting Plate.

Also for the load-displacement testing conducted December 18, 2006, concurrent load and displacement measurements were recorded using a Data Translation model 9804 USB Data Acquisition Function Module and laptop computer.

LOAD-DISPLACEMENT TEST PROCEDURE

In general, the load-displacement tests were conducted using the procedure detailed in the ASTM Specifications. Both the incremental load method and the continuous load method were used. For the incremental load method, after the load and displacement measurement instruments were properly attached to the adhesive anchor, a small load was applied to the system to eliminate any seating displacements in the loading apparatus from affecting the test measurement. Then, an initial load of 500 lbs was applied and held for two minutes. This was followed by applying load at a rate of one stroke from the hand pump every 5 seconds until an approximate magnitude of 2000 lbs was reached. Again, the load was maintained at this level for approximately 2 minutes. After two minutes, the load was increased by approximately 1500 lbs to 3500 lbs at the usual rate of loading and held for another two minutes. This process was repeated by increasing the last pre-pause load level by 1500 lbs until the adhesive anchor was no longer capable of resisting the applied load. Finally, a post failure peak load was attained, if possible, and the rod was removed from the tunnel ceiling using the hydraulic jack.

The continuous load method differed from the incremental method in that from the beginning of the test, load was applied at a rate of one stroke of the hand pump every 5 seconds until the anchor reached its peak load. Load was then again applied at the same rate to attain a post failure peak load, if any. Finally, the hand pump was used to remove the anchor from the ceiling.

LOAD-DISPLACEMENT TEST RESULTS

Incremental Load Method

The peak load data reported for each anchor individually in Appendix A is summarized in Table 6. Plots of the load vs. displacement and time vs. displacement records for each anchor tested are shown in Figures 25 through 28.

Table 6. Summary of Incremental Load-Displacement Test Data

Anchor Label	Sustained Load Test Designation	Initial Protrusion (inches)	Approximate Peak Load/Hold (lbs) / (min.)
WBD084-S-4/5/6-NW	1	2.938	21500 / 0
WBD084-S-7-N	3C	3.313	15000 / 2
WBD084-S-8-N	2	3.031	23000 / 0.6
WBD083-S-4/5/6-NE	6	2.938	15000 / 0.3

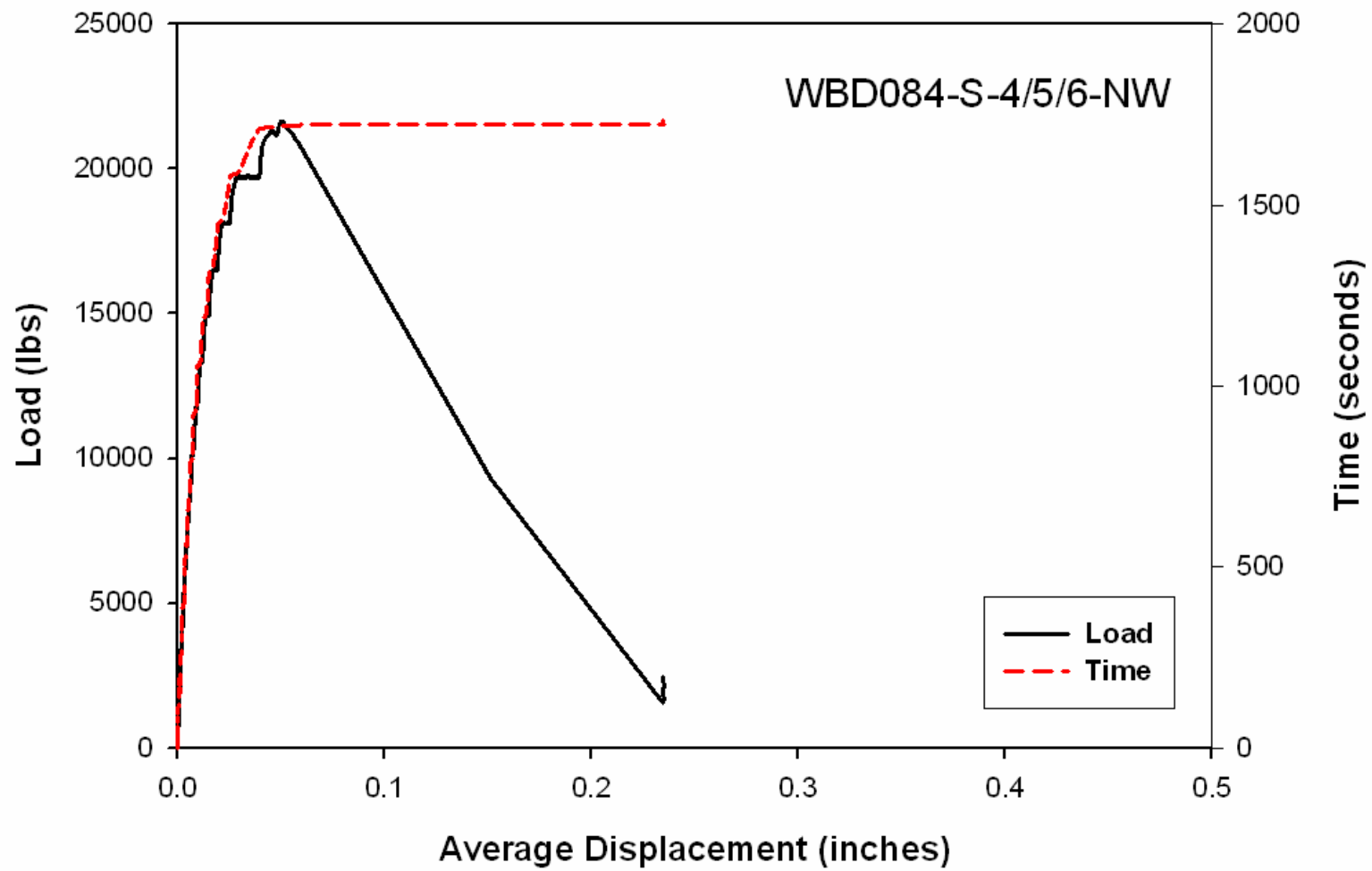


Figure 25. Load and Time versus Displacement Plot for Anchor WBD084-S-4/5/6-NW.

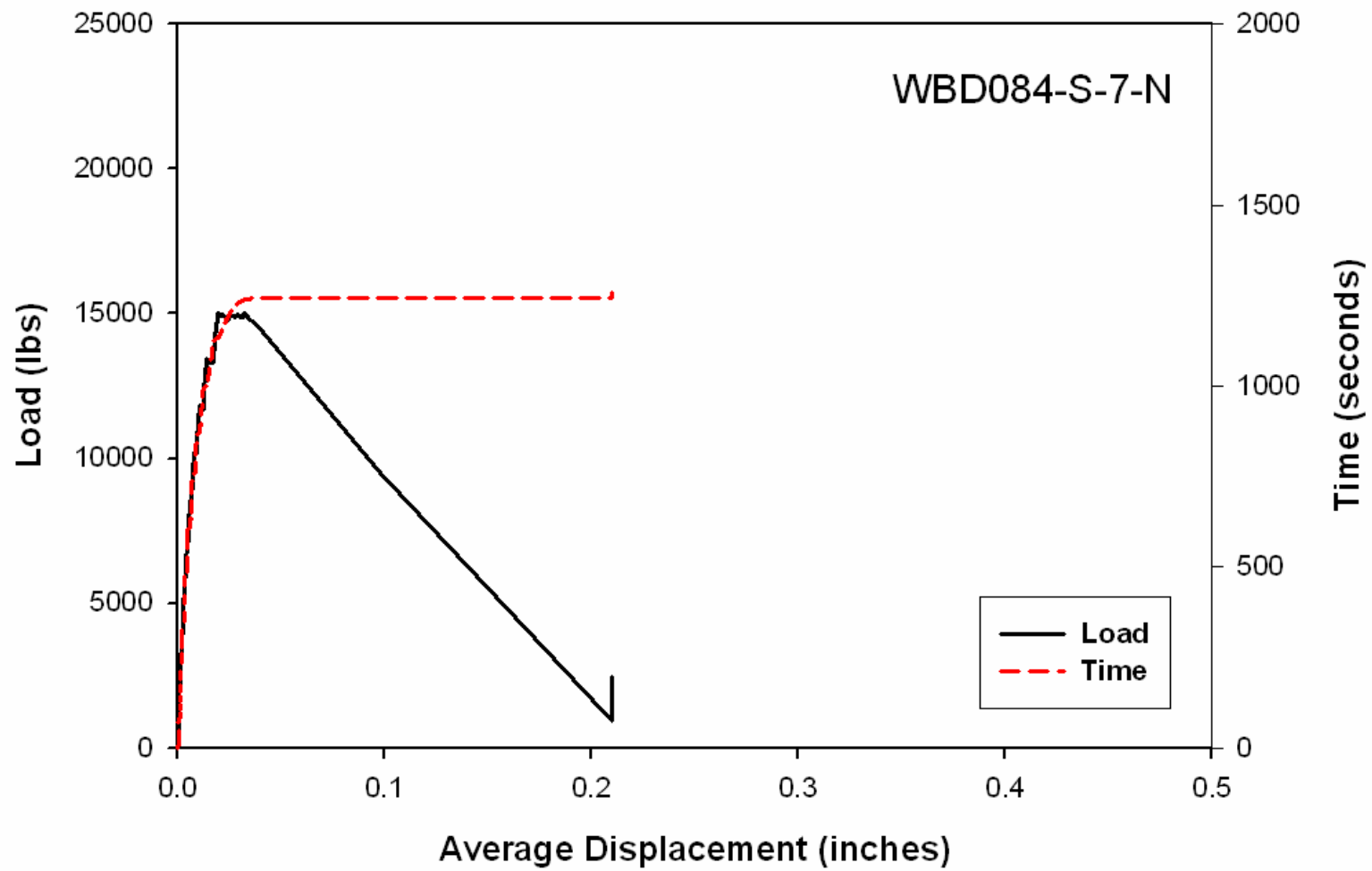


Figure 26. Load and Time versus Displacement Plot for Anchor WBD084-S-7-N.

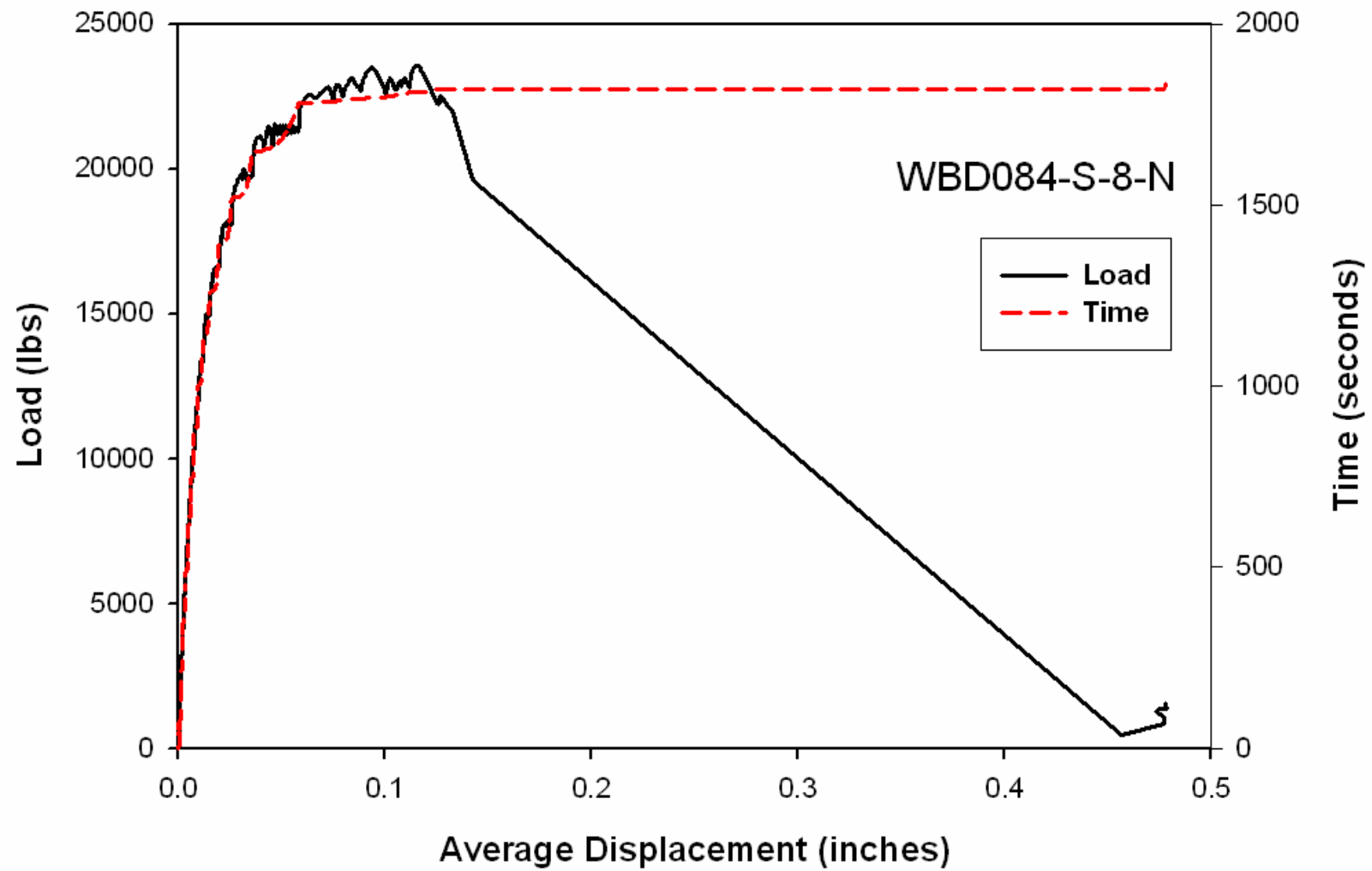


Figure 27. Load and Time versus Displacement Plot for Anchor WBD084-S-8-N.

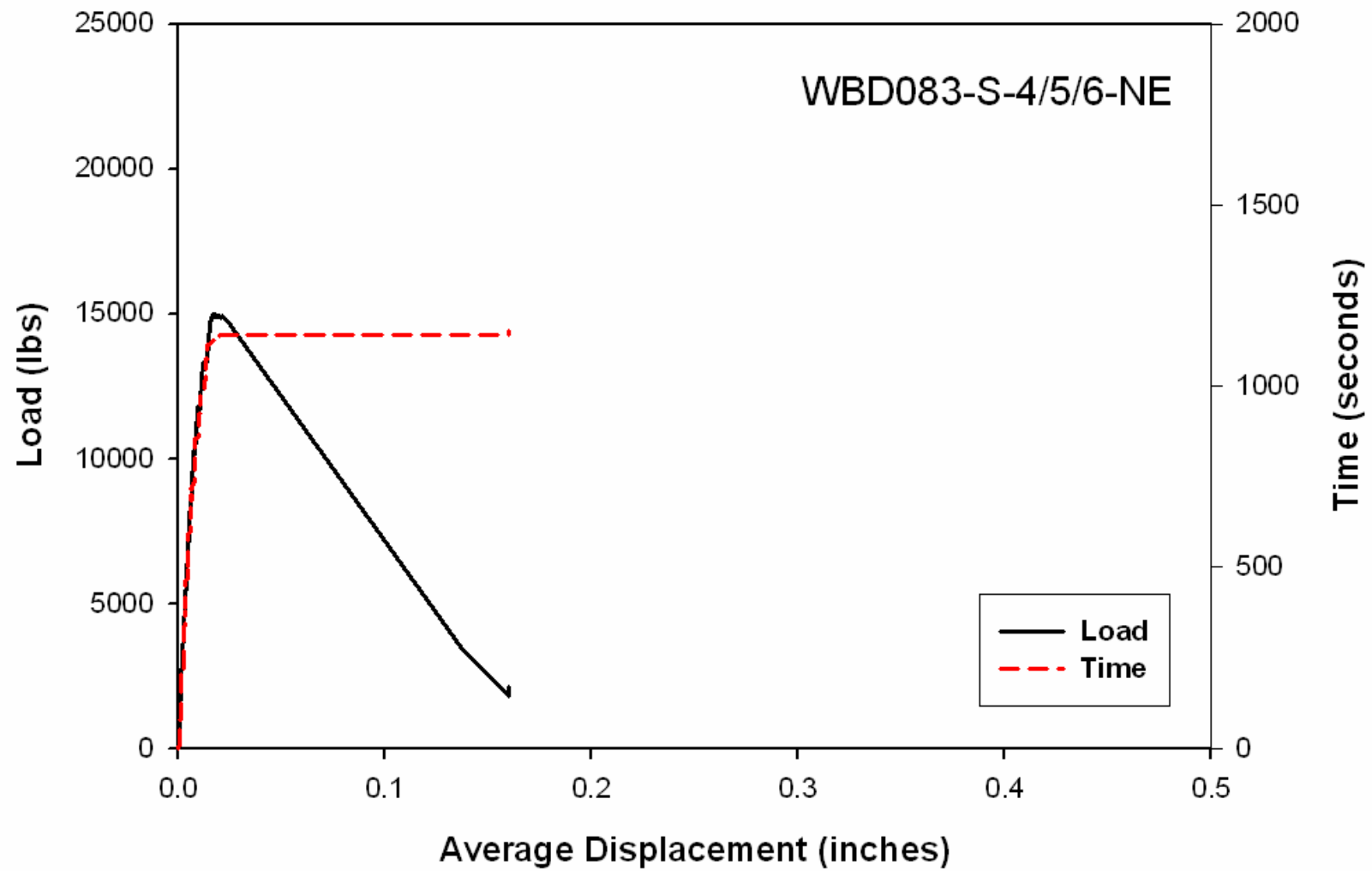


Figure 28. Load and Time versus Displacement Plot for Anchor WBD083-S-4/5/6-NE.

Continuous Load Method

The peak load data reported for each anchor individually in Appendix A is summarized in Table 7. Plots of the load vs. displacement and time vs. displacement records for each anchor tested are shown in Figures 29 through 33.

Table 7. Summary of Continuous Load–Displacement Test Data

Anchor Label	Initial Protrusion (inches)	Peak Load (lbs)	Time to Peak Load (sec.)	Displacement at Peak Load (inches)
WBD084-S-4/5/6-NE	2.938	20257	47	0.035
WBD084-S-4/5/6-SW	2.875	22175	71	0.045
WBD084-S-7-S	3.156	18021	59	0.027
WBD083-S-2-S	3.000	21816	133	0.035
WBD083-S-4/5/6-SW	2.906	22968	88	0.077

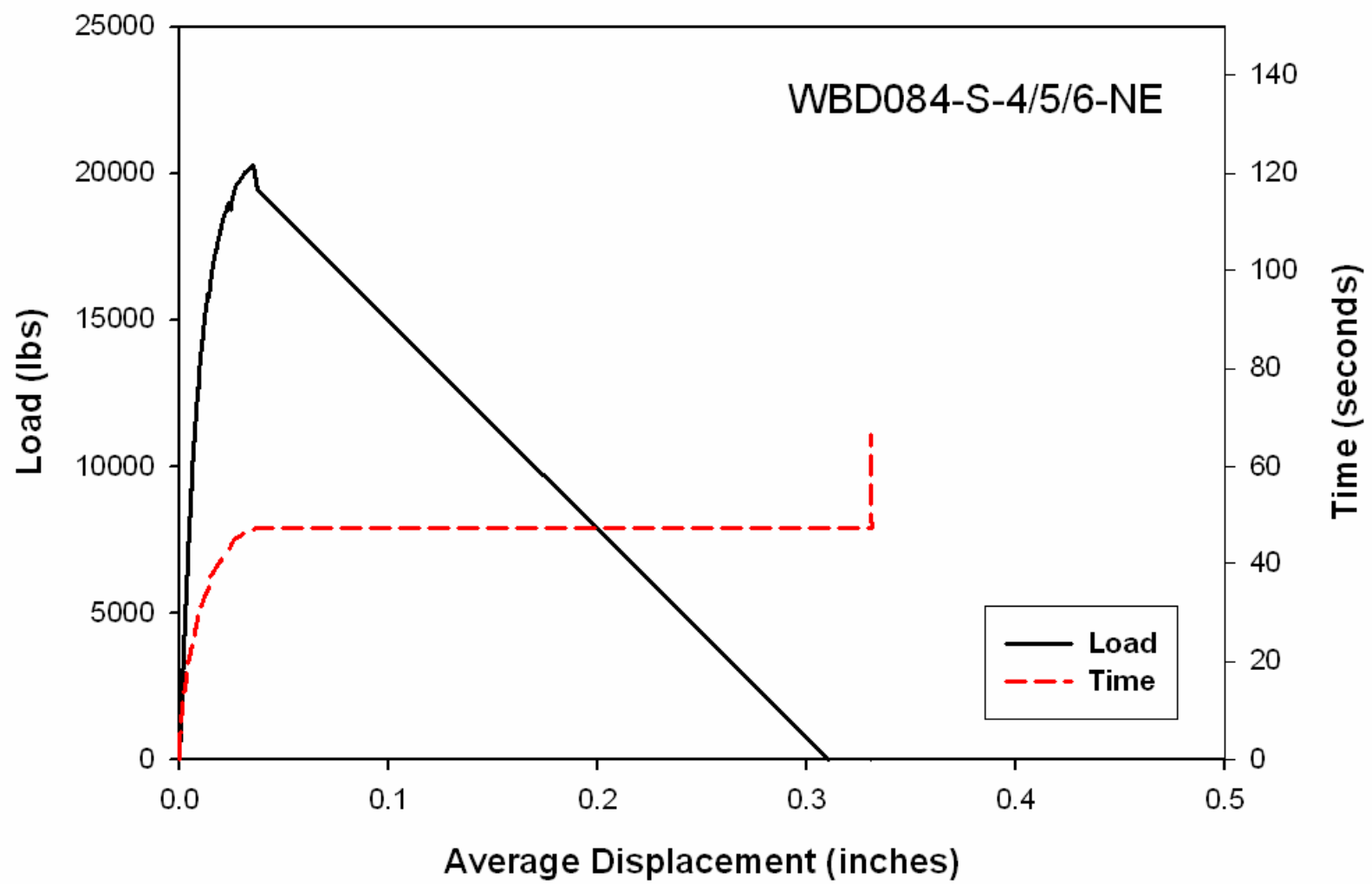


Figure 29. Load and Time versus Displacement Plot for Anchor WBD084-S-4/5/6-NE.

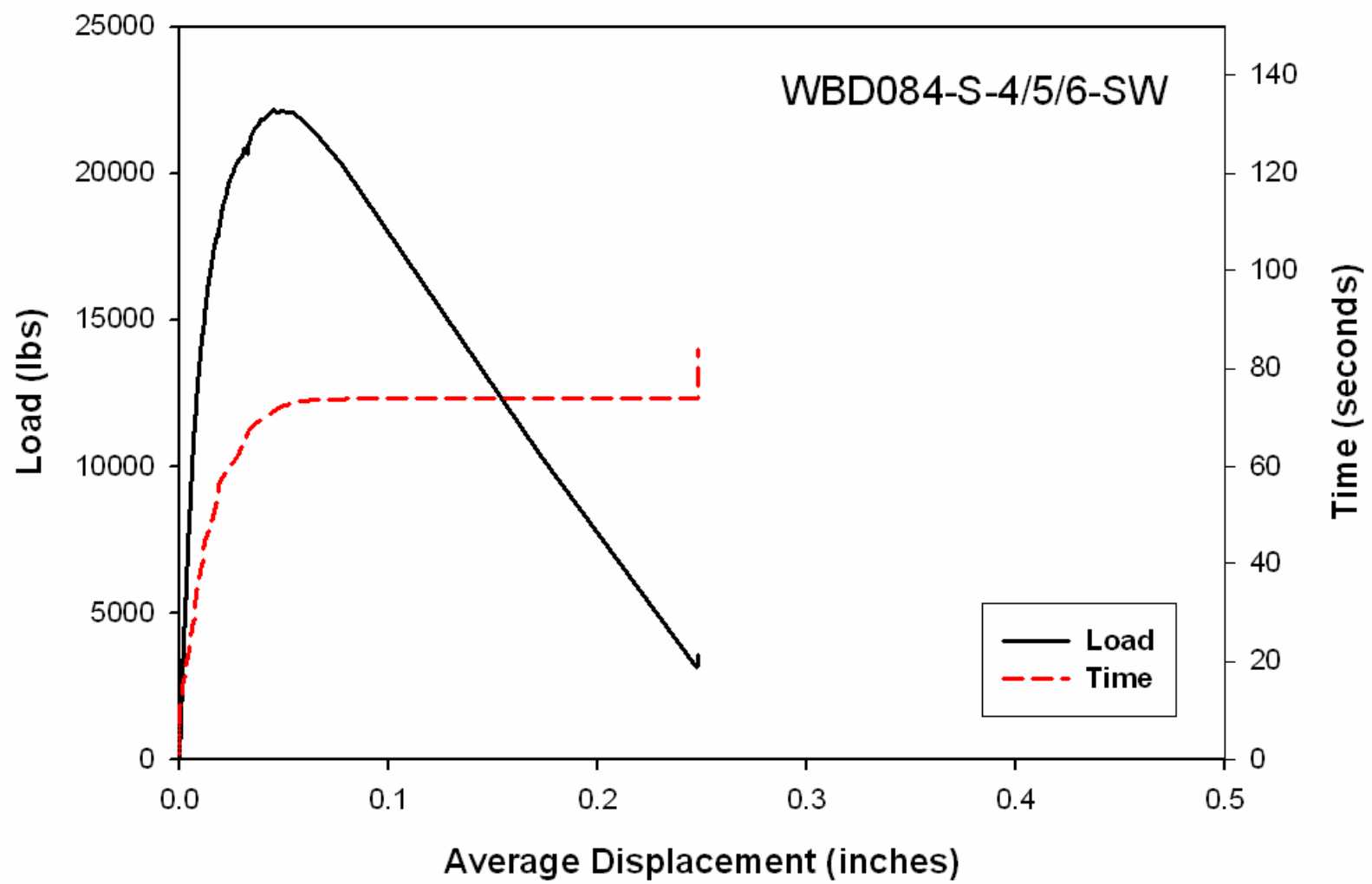


Figure 30. Load and Time versus Displacement Plot for Anchor WBD084-S-4/5/6-SW.

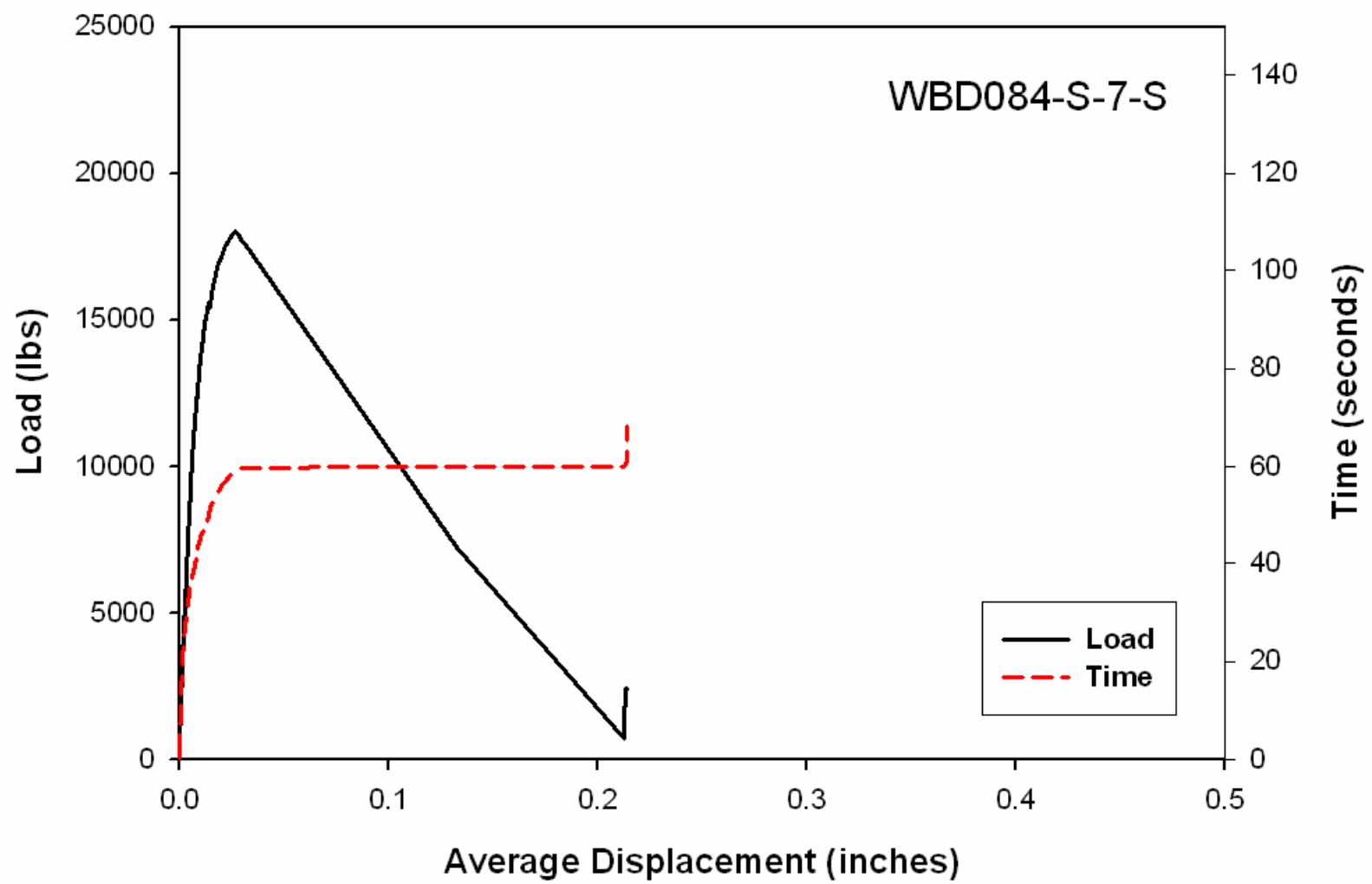


Figure 31. Load and Time versus Displacement Plot for Anchor WBD084-S-7-S.

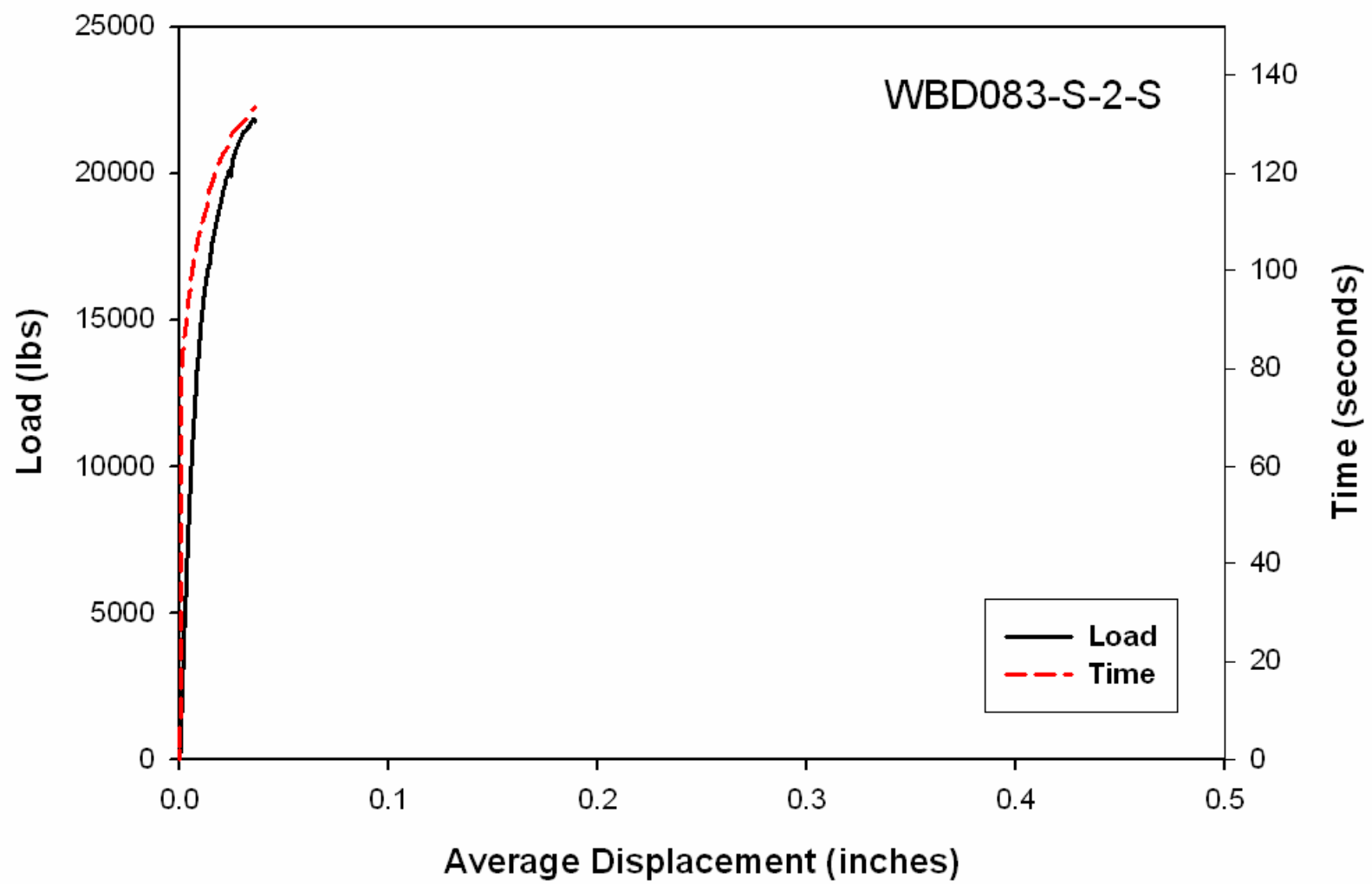


Figure 32. Load and Time versus Displacement Plot for Anchor WBD083-S-2-S.

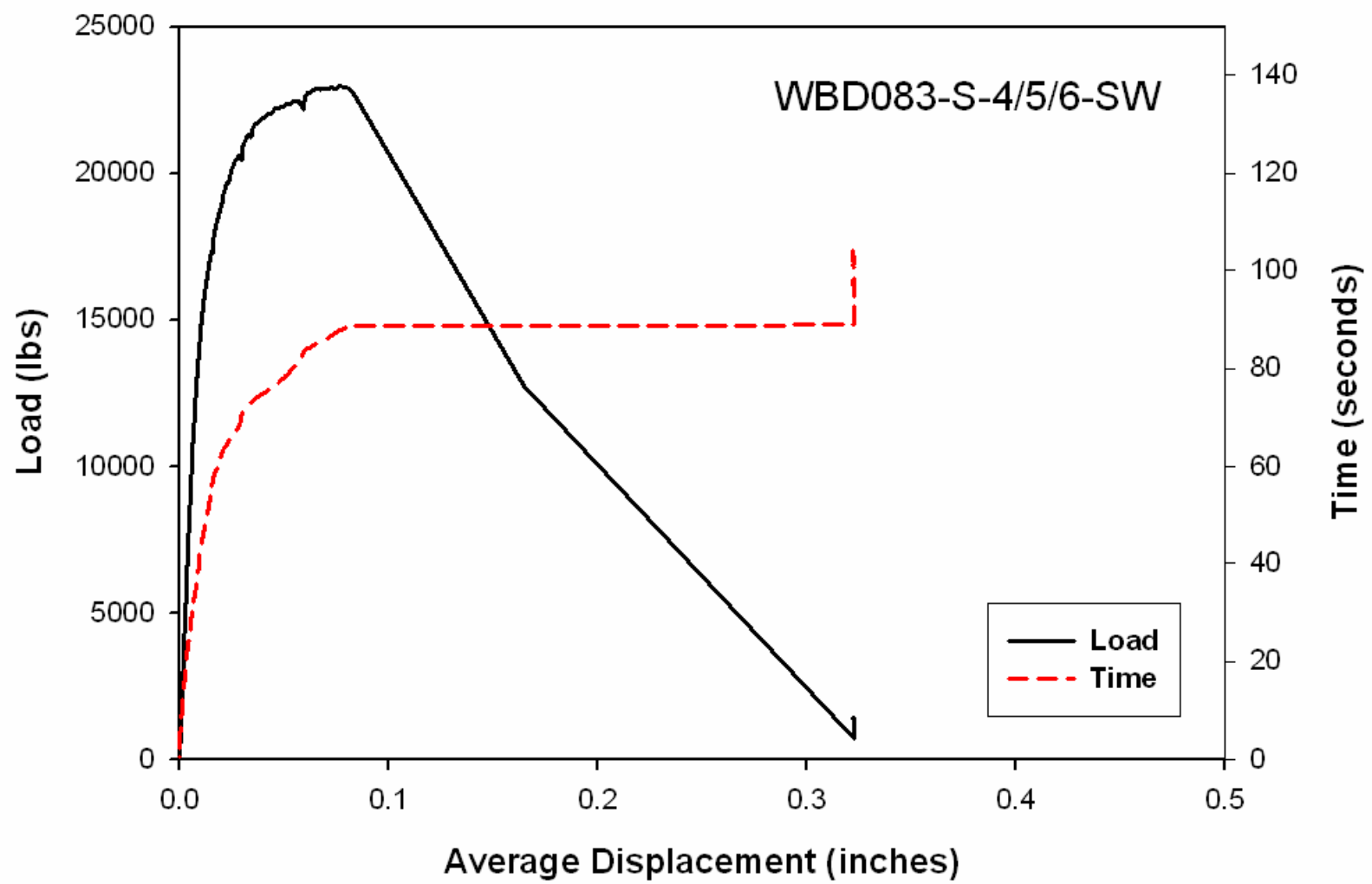


Figure 33. Load and Time versus Displacement Plot for Anchor WBD083-S-4/5/6-SW.

CHAPTER 5. SUMMARY

In support of the National Transportation Safety Board's investigation of the partial collapse of the eastbound I-90 Seaport tunnel ceiling, the Federal Highway Administration conducted sustained load testing on 11 adhesive anchors, and load and load-displacement testing on 15 adhesive anchors in addition to those previously reported.¹ All testing was conducted in full or substantial compliance with the static tension testing provisions of ASTM E 488 *Standard Test Method for Strength of Anchors in Concrete and Masonry Elements*. This document summarized the equipment and procedures used, and reports the data produced during this testing by FHWA.

Of the 9 adhesive anchors subject to the rigorous sustained load investigation, the displacement of 3 anchors never stabilized under the load applied. Two of those anchors displaced completely from the ceiling and the third was relieved of its sustained load once complete displacement was imminent.

In general, each adhesive anchor load or load-displacement tested failed within the epoxy and not at the epoxy-to-concrete interface or, with one exception, spalling of concrete.

While the peak load data from the load tests is reported herein, the results from the sustained load and load-displacement testing are presented graphically due to the quantity of data produced during those experiments. A compact-disk containing the entire time-load-displacement history for each sustained load and load-displacement test has been prepared to accompany this report. The individual anchor test files are contained in a Systat Software Sigmaplot Notebook.

APPENDIX A. TEST SPECIMEN REGISTER

Sustained load screen test initiated July 26, 2006

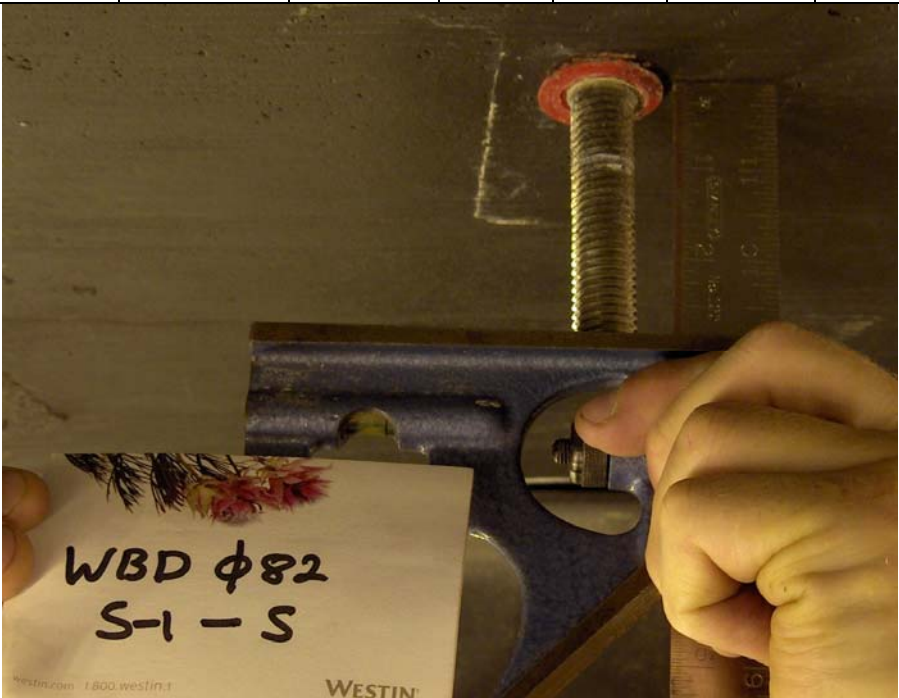
Anchor Label	Test Designation	Initial Protrusion (inches)	Pre-test Displacement		Approx. Sustained Load (lbs)	Duration of Loading (hours)
			East Side (inches)	West Side (inches)		
WBD082-S-1-S	Screen test	3.063	^a	^a	2000	46.5
						
^a – Slight pre-test displacement observed but not quantified.						

Figure 34. Adhesive Anchor WBD082-S-1-S Prior to Sustained Load Test.

Sustained load screen test initiated July 26, 2006

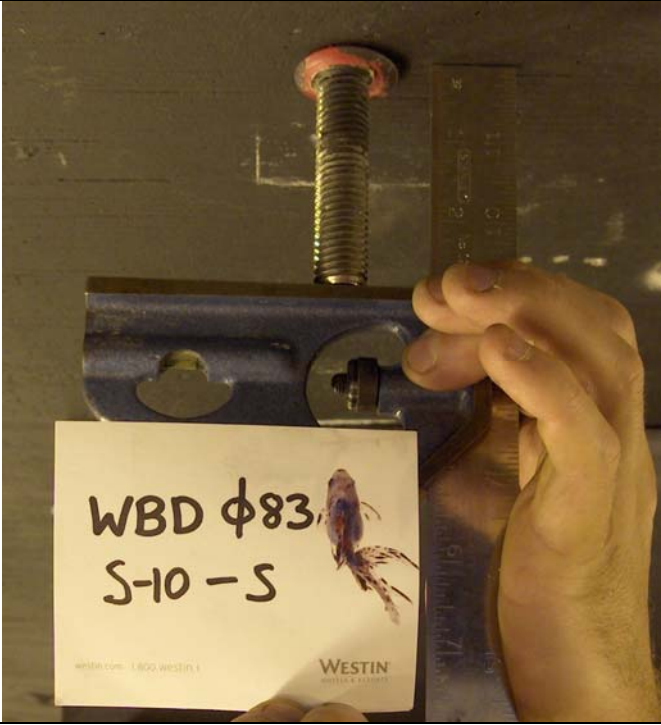
Anchor Label	Test Designation	Initial Protrusion (inches)	Pre-test Displacement		Approx. Sustained Load (lbs)	Duration of Loading (hours)
			East Side (inches)	West Side (inches)		
WBD083-S-10-S	Screen test	2.969			2000	46.5
						

Figure 35. Adhesive Anchor WBD083-S-10-S Prior to Sustained Load Test.

Sustained load test #1 initiated September 26, 2006

Anchor Label	Test Designation	Initial Protrusion (inches)	Pre-test Displacement		Approx. Sustained Load (lbs)	Duration of Loading (hours)
			East Side (inches)	West Side (inches)		
WBD084-S-4/5/6-NW	1	2.906			1000	1508
						

Figure 36. Adhesive Anchor WBD084-S-4/5/6-NW Prior to Sustained Load Test.

Sustained load test #2 initiated September 26, 2006

Anchor Label	Test Designation	Initial Protrusion (inches)	Pre-test Displacement		Approx. Sustained Load (lbs)	Duration of Loading (hours)
			East Side (inches)	West Side (inches)		
WBD084-S-8-N	2	3.000			2000	1508

Figure 37. Adhesive Anchor WBD084-S-8-N Prior to Sustained Load Test.

Sustained load test #3 initiated September 26, 2006

Anchor Label	Test Designation	Initial Protrusion (inches)	Pre-test Displacement		Approx. Sustained Load (lbs)	Duration of Loading (hours)
			East Side (inches)	West Side (inches)		
WBD084-S-9-S	3	3.125			3000	335.5



Figure 37. Adhesive Anchor WBD084-S-9-S Prior to Sustained Load Test.

Sustained load test #4 initiated September 26, 2006

Anchor Label	Test Designation	Initial Protrusion (inches)	Pre-test Displacement		Approx. Sustained Load (lbs)	Duration of Loading (hours)
			East Side (inches)	West Side (inches)		
WBD083-S-1-N	4	2.938			1000	1508

Figure 39. Adhesive Anchor WBD083-S-1-N Prior to Sustained Load Test.

Sustained load test #5 initiated September 26, 2006

Anchor Label	Test Designation	Initial Protrusion (inches)	Pre-test Displacement		Approx. Sustained Load (lbs)	Duration of Loading (hours)
			East Side (inches)	West Side (inches)		
WBD083-S-3-S	5	3.063			2000	75.5

Figure 40. Adhesive Anchor WBD083-S-3-S Prior to Sustained Load Test.

Sustained load test #6 initiated September 26, 2006

Anchor Label	Test Designation	Initial Protrusion (inches)	Pre-test Displacement		Approx. Sustained Load (lbs)	Duration of Loading (hours)
			East Side (inches)	West Side (inches)		
WBD083-S-4/5/6-NE	6	2.938			3000	1508



Figure 41. Adhesive Anchor WBD083-S-4/5/6-NE Prior to Sustained Load Test.

Sustained load test #3B initiated October 12, 2006

Anchor Label	Test Designation	Initial Protrusion (inches)	Pre-test Displacement		Approx. Sustained Load (lbs)	Duration of Loading (hours)
			East Side (inches)	West Side (inches)		
WBD083-S-2-N	3B	2.938			3000	7

Figure 42. Adhesive Anchor WBD083-S-2-N Prior to Sustained Load Test.

Sustained load test #5B initiated October 12, 2006

Anchor Label	Test Designation	Initial Protrusion (inches)	Pre-test Displacement		Approx. Sustained Load (lbs)	Duration of Loading (hours)
			East Side (inches)	West Side (inches)		
WBD083-S-3-N	5B	3.094			2000	1130.5

Figure 43. Adhesive Anchor WBD083-S-3-N Prior to Sustained Load Test.

Sustained load test #3C initiated October 26, 2006.

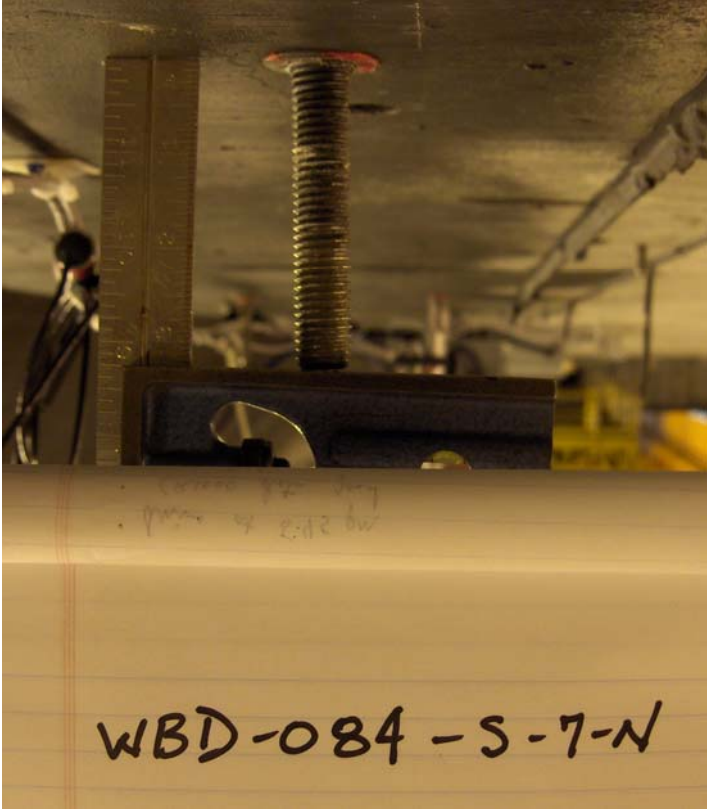
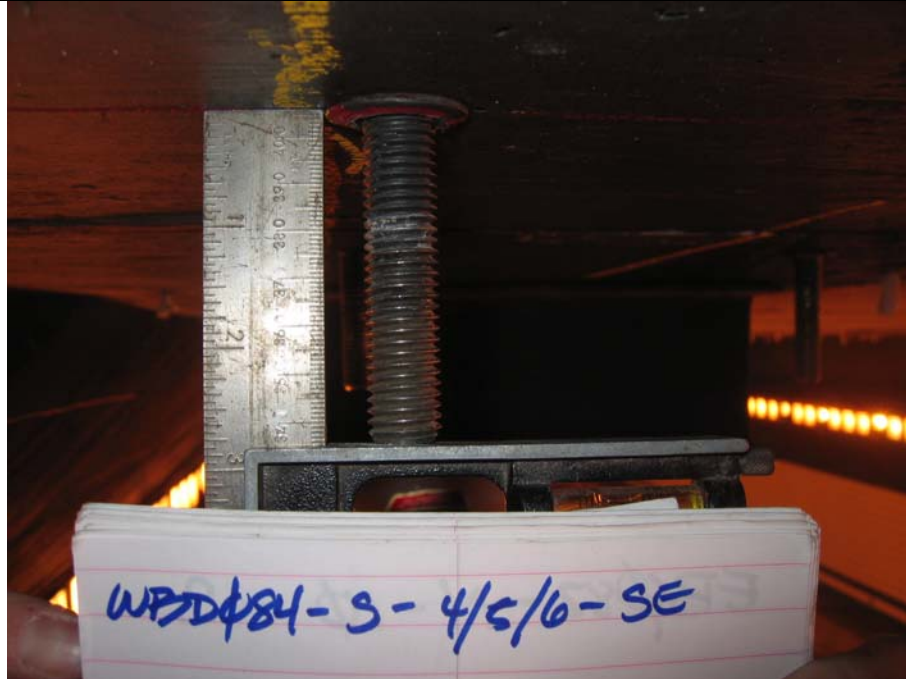
Anchor Label	Test Designation	Initial Protrusion (inches)	Pre-test Displacement		Approx. Sustained Load (lbs)	Duration of Loading (hours)
			East Side (inches)	West Side (inches)		
WBD084-S-7-N	3C	3.250			3000	814
						

Figure 44. Adhesive Anchor WBD084-S-7-N Prior to Sustained Load Test.

Anchor 1 of 6 load tested December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load ^a (lbs)	Time to Peak Load (sec.)
		East Side (inches)	West Side (inches)		
WBD084-S-4/5/6-SE	2.813			24389	80



^a – Represents the adjusted load based on load cell calibration, see Appendix B.

Figure 45. Adhesive Anchor WBD084-S-4/5/6-SE Prior to Load Test.

Anchor 2 of 6 load tested December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load ^a (lbs)	Time to Peak Load (sec.)
		East Side (inches)	West Side (inches)		
WBD084-S-8-S	3.531	0.250	0.281	15519	80



^a – Represents the adjusted load based on load cell calibration, see Appendix B.

Figure 46. Adhesive Anchor WBD084-S-8-S Prior to Load Test.

Anchor 3 of 6 load tested December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load ^a (lbs)	Time to Peak Load (sec.)
		East Side (inches)	West Side (inches)		
WBD083-S-4/5/6-NW	3.000	^b	^b	21688	53

^a – Represents the adjusted load based on load cell calibration, see Appendix B.
^b – Slight pre-test displacement observed but not quantified.

Figure 47. Adhesive Anchor WBD083-S-4/5/6-NW Prior to Load Test.

Anchor 4 of 6 load tested December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load ^a (lbs)	Time to Peak Load (sec.)
		East Side (inches)	West Side (inches)		
WBD083-S-4/5/6-SE	2.844	^b	^b	21819	50

^a – Represents the adjusted load based on load cell calibration, see Appendix B.
^b – Slight pre-test displacement observed but not quantified.

Figure 48. Adhesive Anchor WBD083-S-4/5/6-SE Prior to Load Test.

Anchor 5 of 6 load tested December 18, 2006.

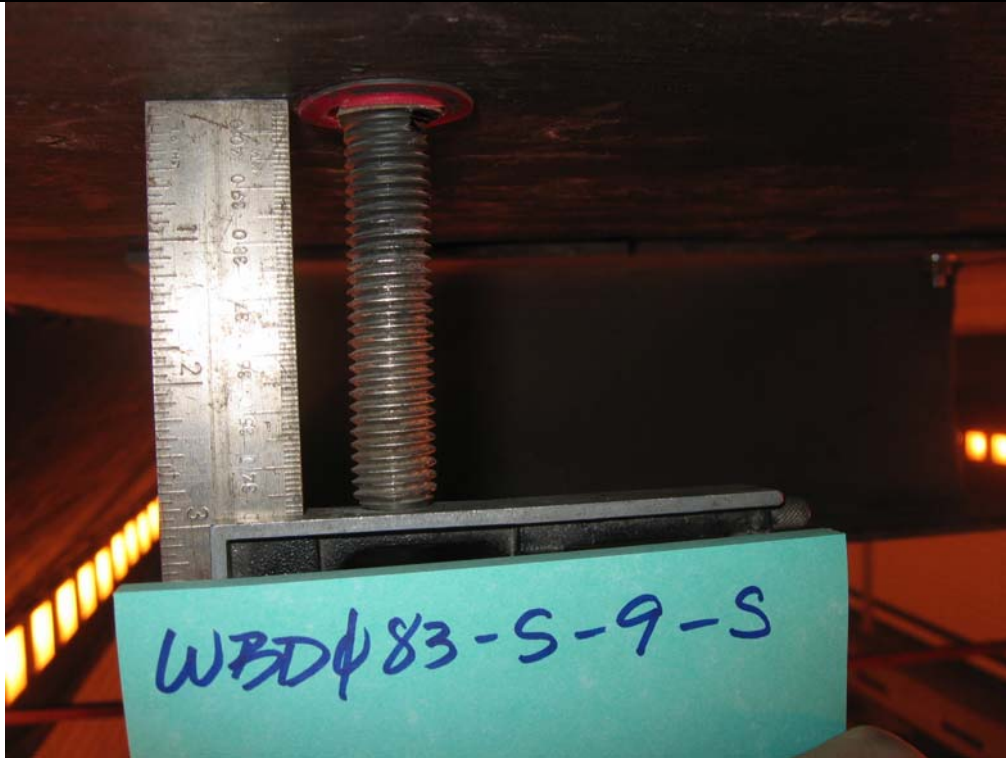
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load ^a (lbs)	Time to Peak Load (sec.)
		East Side (inches)	West Side (inches)		
WBD083-S-9-N	2.875	^b	^b	17460	52

^a – Represents the adjusted load based on load cell calibration, see Appendix B.
^b – Slight pre-test displacement observed but not quantified.

Figure 49. Adhesive Anchor WBD083-S-9-N Prior to Load Test.

Anchor 6 of 6 load tested December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load ^a (lbs)	Time to Peak Load (sec.)
		East Side (inches)	West Side (inches)		
WBD083-S-9-S	2.938			23374	77



^a – Represents the adjusted load based on load cell calibration, see Appendix B.

Figure 50. Adhesive Anchor WBD083-S-9-S Prior to Load Test.

Anchor 1 of 4 load–displacement tested with the incremental method on December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load ^a / Hold (lbs) / (min.)
		East Side (inches)	West Side (inches)	
WBD084-S-4/5/6-NW	2.938			21500 / 0



^a – Represents the adjusted load based on load cell calibration, see Appendix B.

Figure 51. Adhesive Anchor WBD084-S-4/5/6-NW Prior to Load–Displacement Test.

Anchor 2 of 4 load–displacement tested with the incremental method on December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load ^a / Hold (lbs) / (min.)
		East Side (inches)	West Side (inches)	
WBD084-S-7-N	3.313	^b	^b	15000 / 2

^a – Represents the adjusted load based on load cell calibration, see Appendix B.
^b – Slight pre-test displacement observed but not quantified.

Figure 52. Adhesive Anchor WBD084-S-7-N Prior to Load–Displacement Test.

Anchor 3 of 4 load–displacement tested with the incremental method on December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load ^a / Hold (lbs) / (min.)
		East Side (inches)	West Side (inches)	
WBD084-S-8-N	3.063	^b	^b	22000 / 1

^a – Represents the adjusted load based on load cell calibration, see Appendix B.
^b – Slight pre-test displacement observed but not quantified.

Figure 53. Adhesive Anchor WBD084-S-8-N Prior to Load–Displacement Test.

Anchor 4 of 4 load–displacement tested with the incremental method on December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load ^a / Hold (lbs) / (min.)
		East Side (inches)	West Side (inches)	
WBD083-S-4/5/6-NE	2.938	0.000	0.188	14000 / 0

^a – Represents the adjusted load based on load cell calibration, see Appendix B.

Figure 54. Adhesive Anchor WBD083-S-4/5/6-NE Prior to Load–Displacement Test.

Anchor 1 of 5 load–displacement tested with the continuous method on December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load ^a (lbs)	Time to Peak Load (sec.)	Displacement at Peak Load (inches)
		East Side (inches)	West Side (inches)			
WBD084-S-4/5/6-NE	2.938			20257	47	0.035

^a – Represents the adjusted load based on load cell calibration, see Appendix B.

Figure 55. Adhesive Anchor WBD084-S-4/5/6-NE Prior to Load–Displacement Test.

Anchor 2 of 5 load–displacement tested with the continuous method on December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load ^a (lbs)	Time to Peak Load (sec.)	Displacement at Peak Load (inches)
		East Side (inches)	West Side (inches)			
WBD084-S-4/5/6-SW	2.875			22175	71	0.045
						
^a – Represents the adjusted load based on load cell calibration, see Appendix B.						

Figure 56. Adhesive Anchor WBD084-S-4/5/6-SW Prior to Load–Displacement Test.

Anchor 3 of 5 load–displacement tested with the continuous method on December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load ^a (lbs)	Time to Peak Load (sec.)	Displacement at Peak Load (inches)
		East Side (inches)	West Side (inches)			
WBD084-S-7-S	3.156			18021	59	0.027

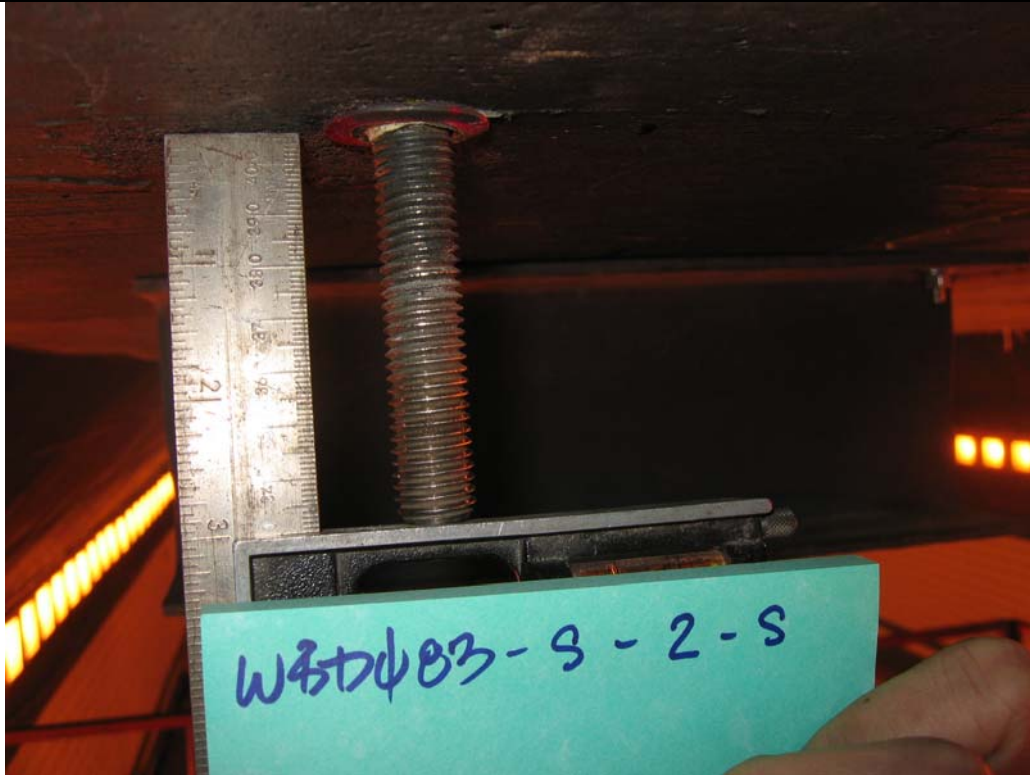


^a – Represents the adjusted load based on load cell calibration, see Appendix B.

Figure 57. Adhesive Anchor WBD084-S-7-S Prior to Load–Displacement Test.

Anchor 4 of 5 load–displacement tested with the continuous method on December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load ^a (lbs)	Time to Peak Load (sec.)	Displacement at Peak Load (inches)
		East Side (inches)	West Side (inches)			
WBD083-S-2-S ^b	3.000			21816	133	0.035



- ^a – Represents the adjusted load based on load cell calibration, see Appendix B.
^b – Shallow cone failure.

Figure 58. Adhesive Anchor WBD083-S-2-S Prior to Load–Displacement Test.

Anchor 5 of 5 load–displacement tested with the continuous method on December 18, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load ^a (lbs)	Time to Peak Load (sec.)	Displacement at Peak Load (inches)
		East Side (inches)	West Side (inches)			
WBD083-S-4/5/6-SW	2.906			22968	88	0.077



^a – Represents the adjusted load based on load cell calibration, see Appendix B.

Figure 59. Adhesive Anchor WBD083-S-4/5/6-SW Prior to Load–Displacement Test.

APPENDIX B. LOAD CELL CALIBRATION

Load measurements during December 18, 2006 were made with Wagner Instruments (WI) 50 kip load cell S/N: 3344. A formal load cell calibration against a certified load standard was performed prior to testing on December 15, 2006 and subsequent to testing on December 21, 2006. The December 15, 2006 calibration was conducted by reacting the Wagner load cell against a load cell calibrated yearly to a NIST load standard. The results, shown in Figure 60, indicate essentially ideal linear behavior.

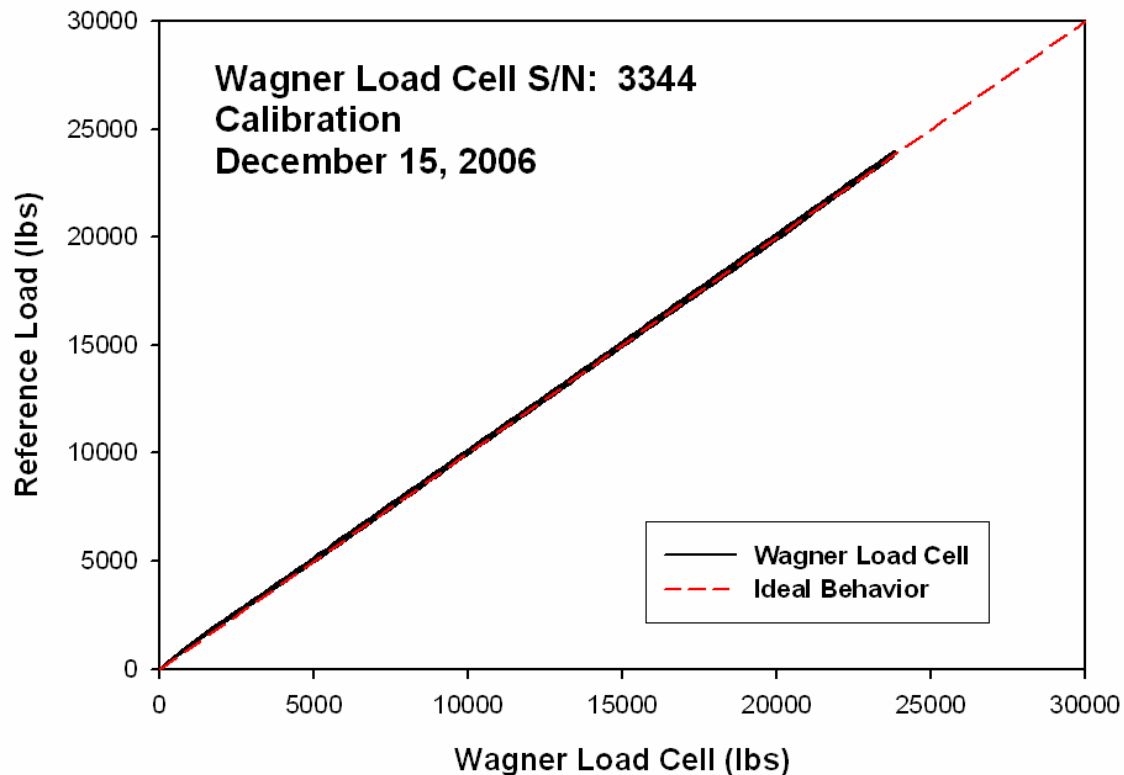


Figure 60. December 15, 2006 Load Cell Calibration.

The calibration conducted December 21, 2006 utilized the same reference load cell. However, during this calibration, load was applied to the Wagner load cell through the identical series of plates and shims used on-site during testing to more accurately represent the actual boundary conditions experienced. The results, shown in Figure 61, are slightly non-linear and represent an approximate 6% amplification of load. Therefore, loads measured during the load and load-displacement testing conducted December 18, 2006 and detailed in this report have been reduced by applying a multiplier of 0.94.

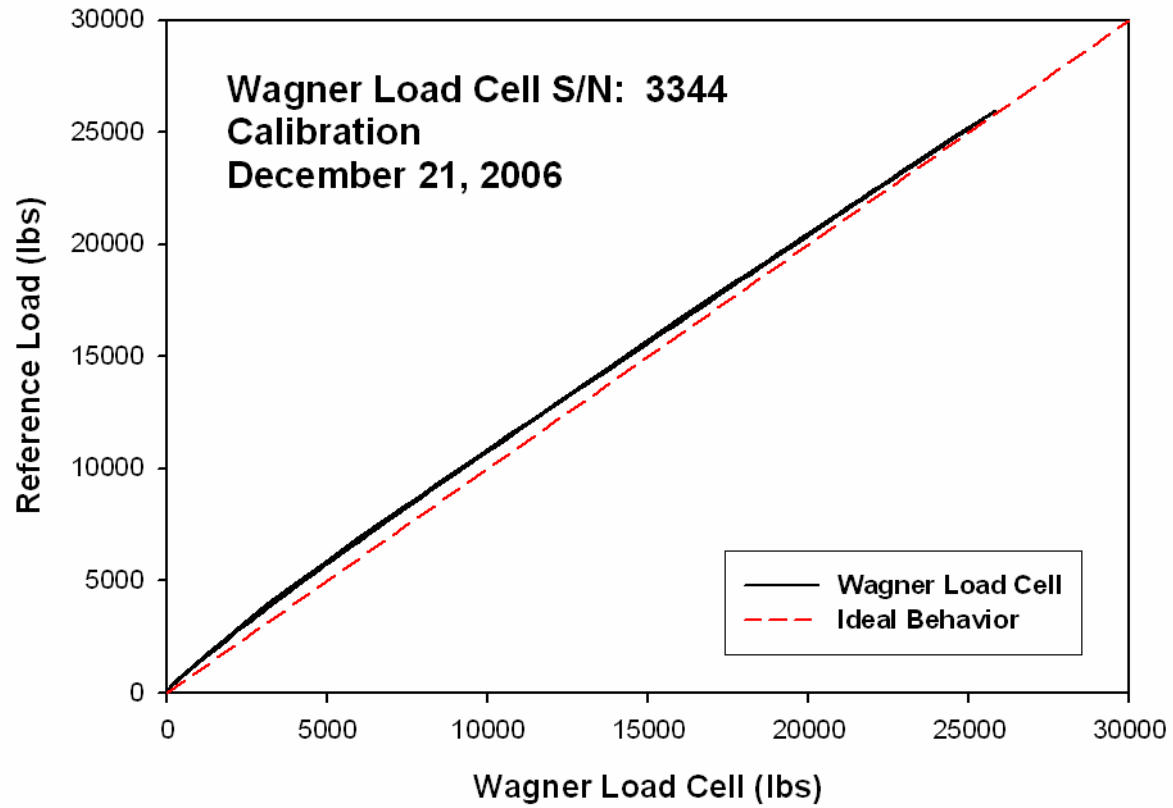


Figure 61. December 21, 2006 Load Cell Calibration.